

Chapter 17 was amended by [Local Law 77 of 2023](#). This law has an effective date of June 10, 2023.

CHAPTER 17

SPECIAL INSPECTIONS AND TESTS

SECTION BC 1701 GENERAL

1701.1 Scope. The provisions of this chapter shall govern the inspection of quality, workmanship and requirements for construction. Materials, inspection and testing shall conform to the applicable standards listed in this code or in the rules of the department. See Chapter 1 of Title 28 of the *Administrative Code* for additional provisions relating to materials, testing and inspections.

1701.2 New materials. See Chapter 1 of Title 28 of the *Administrative Code* for additional provisions.

1701.3 Used materials. See Chapter 1 of Title 28 of the *Administrative Code*.

SECTION BC 1702 DEFINITIONS

1702.1 Definitions. The following terms are defined in Chapter 2:

APPROVED AGENCY.

APPROVED FABRICATOR.

CERTIFICATE OF COMPLIANCE.

FABRICATED ITEM.

INSPECTION CERTIFICATE.

INTUMESCENT FIRE-RESISTANT COATINGS.

LABEL.

MANUFACTURER'S DESIGNATION.

MARK.

MASTIC FIRE-RESISTANT COATINGS.

SPECIAL INSPECTION.

SPECIAL INSPECTION, CONTINUOUS.

SPECIAL INSPECTION, PERIODIC.

SPECIAL INSPECTOR.

SPRAYED FIRE-RESISTANT MATERIALS.

SUPERINTENDENT OF CONSTRUCTION.

**SECTION BC 1703
APPROVALS**

1703.1 Approved Agency.[‡] See Chapter 1 of Title 28 of the *Administrative Code*.

**SECTION BC 1704
RESPONSIBILITIES FOR SPECIAL
INSPECTIONS AND TESTS**

1704.1 General. Where application is made for construction as described in this section, one or more special inspection agencies meeting the requirements of Sections 28-114.1 and 28-115.1 of the *Administrative Code* shall be retained by the owner to provide inspections during construction on the types of work listed under Section 1705 and elsewhere in this code. These inspections are in addition to the other inspections required by Section 110, elsewhere in this code, and Section 28-116 of the *Administrative Code*. The special inspector shall be acceptable to the registered design professional of record.

Exception: Special inspections are not required for building components unless the design involves the practice of professional engineering or architecture as defined by the *Education Law of the State of New York* and applicable regulations governing the professional registration and certification of engineers or architects.

1704.1.1 Responsibilities. Special inspection responsibilities of registered design professionals of record, special inspection agencies, and permit holders shall meet the applicable requirements of this section.

1704.1.1.1 Responsibilities of the registered design professional of record. The registered design professional of record shall have the following duties and responsibilities:

- 1. Identification of required special inspections.** The registered design professional of record shall identify the materials, equipment, installation, fabrication, erection or placement of components and connections, or construction operations for the work indicated on the construction documents that are subject to special inspections on the construction documents and in any other form supplied or manner required by the department prior to the approval of the construction documents.
- 2. Respond to field discrepancies.** The registered design professional of record shall respond to special inspector reports of uncorrected discrepancies and shall approve remedial measures.
- 3. Review of shop drawings and other submittals.** Where shop or working drawings, and other submittals have been provided, the registered design professional of record shall review such shop drawings for conformance with the approved design. The registered design professional shall acknowledge conformance with the approved design on such documents.
- 4. Deviations from approved construction documents.** The registered design professional of record shall submit to the department and the special inspection agency written documentation of the professional's approval of deviations from the approved construction documents. All such deviations approved by the registered design professional of record shall be filed with the department on amended construction documents in accordance with Section 28-104.3 of the *Administrative Code*.

1704.1.1.2 Responsibilities of the special inspection agency. The special inspection agency shall have the following duties and responsibilities:

- 1. Independence.** The special inspection agency shall be independent of the contractors responsible for the work being inspected. The registered design professional of record is permitted to act as the approved special inspection agency and such agency's personnel are permitted to act as special inspectors for the work designed by the registered design professional of record, provided those personnel meet the qualification requirements of this section to the satisfaction of the department.
- 2. Qualifications.** The special inspection agency shall ensure that qualified special inspectors are provided to perform special inspections in accordance with the rules of the department. Upon request of the registered design professional of record, the special inspection agency shall provide proof of inspector's and agency's qualifications, certifications, and other documents demonstrating compliance with department rules and regulations.

3. **Construction documents.** The special inspection agency shall examine all approved construction documents that relate to the work that is the subject of the special inspections to confirm that the documents are sufficient to enable the proper performance of the special inspection. The special inspection agency shall confirm that any shop drawings or sketches have been accepted by the registered design professional of record. Acceptance shall be demonstrated in writing by the registered design professional of record on the documents. All approved shop drawings, submittals and/or sketches are to be located on site and accessible to the special inspector at time of inspection.
4. **Inspection.** The special inspection agency shall observe work subject to special inspection to confirm that the work that is the subject of the special inspection is in compliance with the approved construction documents; with the approved shop drawings, where provided; and with the special inspection requirements of this code and department rules and regulations.
5. **Tests.** Where specifically required by this code, the performance of testing shall be by qualified special inspectors working for the special inspection agency. In all other instances, the special inspection agency shall observe testing to confirm that testing activities that are subject to special inspection are in compliance with the prescribed testing procedures.
6. **Reports.** The special inspection agency shall prepare reports of its inspections and tests, and such reports shall indicate that work inspected was or was not performed in conformance with approved construction documents.
 - 6.1. **Format.** Special inspection reports and records of special inspections shall be in the form and format supplied by the commissioner.
 - 6.2. **Discrepancies.** Discrepancies shall be brought to the immediate attention of the contractor and, when applicable, to the attention of the superintendent of construction, for correction. The discrepancies shall also be brought to the attention of the owner, and the registered design professional of record in a timely manner to allow for correction of the discrepancies.
 - 6.3. **Hazardous conditions.** The special inspector shall report conditions noted as hazardous to life, safety or health that are not immediately corrected to the immediate attention of the commissioner.
 - 6.4. **Final report.** Prior to issuance of a certificate of occupancy or letter of completion, a final report documenting the required special inspections, including any required performance or witnessing of tests, and correction of all discrepancies noted in the inspections shall be submitted to the department. Such report shall certify compliance with approved construction documents, deviations approved by the registered design professional of record in accordance with Section 1704.1.1.1 Item 4 of this code, and the administrative and procedural requirements of this code.
 - 6.5. **Approval of partially completed work.** Reports of partially completed work shall be accepted when such reports indicate that the inspected work has been completed in accordance with the construction documents, and indicate the extent and the condition of the remaining work.
 - 6.6. **Maintain records.** The special inspection agency shall maintain records of inspections for a period of 6 years from the date of the special inspection agency's final submittal or report of completed inspection. Such records shall be supplied to the commissioner upon request. The commissioner may require that special inspection reports be filed with the department and/or that such reports be otherwise made accessible for review.

1704.1.1.3 Responsibilities of the permit holder. The permit holder shall have the following duties and responsibilities:

1. **Statement of special inspections.** The application for permit shall include a statement of relevant special inspections as a condition for permit issuance on forms supplied by the department. This statement shall include:
 - 1.1. A complete list of materials and work included in the permit application requiring special inspections by this section and elsewhere in this code and the inspections to be performed on such work and materials; and
 - 1.2. A list of the special inspection agencies retained by the owner for conducting the special inspections listed on the approved plans. Within each category of work, multiple special inspectors or special inspection agencies may be employed.

SPECIAL INSPECTIONS AND TESTS

- 2. Identification of registered design professionals providing design documents for certain construction operations.** The registered design professionals responsible for the production of design, sequence of construction operations or shop drawings for projects that require design pursuant to Chapter 33, shall file documentation of their intent to perform those duties on forms supplied by the department. The registered design professionals preparing such documents shall also be subject to the responsibilities of a registered design professional of record required by Section 1704.1.1.1 for the scope of work covered in the design, sequence of construction operations or shop drawings for which they are responsible.
- 3. Required notification to special inspection agencies.** The holder of a permit covering work requiring a special inspection shall notify the relevant special inspectors or special inspection agencies in writing at least 72 hours before the commencement of any work requiring special inspection. The permit holder shall make available the most current approved construction documents to the special inspection agency.
- 4. Access for special inspection.** The construction or work for which special inspection is required shall remain accessible and exposed for special inspection purposes until completion of the required special inspections.
- 5. Resolution of discrepancies.** The permit holder shall be responsible for the resolution of identified discrepancies. A satisfactory inspection by a special inspection agency or the acceptance by the department of a satisfactory report of an inspection by a special inspection agency shall not be construed to be an approval by the department of a violation of the provisions of this code or of any other provision of law.

1704.2 Inspection of fabricators. Where fabrication of structural members, and other regulated products is performed on the premises of a fabricator's shop, special inspection and progress inspection of the fabricated items shall be required by this section, the *New York City Construction Codes*, and Chapter 1 of Title 28 of the *Administrative Code*.

1704.2.1 Fabrication and implementation procedures. The special inspector shall verify that the fabricator maintains detailed fabrication and quality control procedures that provide a basis for inspection control of the workmanship and the fabricator's ability to conform to approved construction documents and referenced standards. The special inspector shall review the procedures for completeness and adequacy relative to the code requirements for the fabricator's scope of work.

Exception: Special inspections and progress inspections as required by Section 1704.2 shall not be required where the fabricator is approved in accordance with Chapter 1 of Title 28 of the *Administrative Code*, except as required by Section 1704.2.2 of this code.

1704.2.2 Fabricator approval. Work that is subject to special inspection and performed on the approved fabricator's premises shall be inspected by the special inspection agency in accordance with Section 1704.2.2.3. Approval of fabricators shall be based upon review of the fabricator's written procedural and quality control manuals and periodic auditing of fabrication practices by an approved agency.

1704.2.2.1 Certificate of intent to fabricate. For all work where approval is sought based upon Section 1704.2.2, the approved fabricator shall submit a certificate of intent to fabricate such work to the department identifying the work to be performed as an approved fabricator, and that such work shall be performed in accordance with the approved construction documents, referenced standards and applicable provisions of law and for which site the work is being fabricated prior to fabricating any items for such project.

1704.2.2.2 Fabricator's certificate of compliance. For all fabricated items, the approved fabricator shall submit a certificate of compliance to the department stating that the work was performed in accordance with the approved construction documents, referenced standards and applicable provisions of law.

1704.2.2.3 In-plant special inspections. The owner shall engage an approved special inspection agency to perform special inspections for the portion of the work performed in the approved fabricator's premises. These inspections shall at a minimum include: (i) one visit to the fabricator's premises during the course of the subject work to verify the fabricator's compliance with the written and procedural quality control manuals; and (ii) inspection of 5 percent of the subject work to verify compliance with the approved construction documents.

SECTION BC 1705 REQUIRED SPECIAL INSPECTIONS AND TESTS

1705.1 General. Special inspections and tests of structural elements and nonstructural components of buildings and structures shall meet the applicable requirements of this section.

1705.1.1 Special cases. Special inspections shall be required for proposed work that is, in the opinion of the commissioner, unusual in its nature, including, but not limited to:

1. Construction materials and systems that are alternatives to materials and systems prescribed by this code.
2. Unusual design applications of materials described in this code.
3. Materials and systems required to be installed in accordance with manufacturer's instructions that prescribe additional requirements not contained in this code or in standards referenced by this code.

1705.1.1.1 Qualifications. Where special inspections are required for proposed work in accordance with Section 1705.1.1, the special inspection shall be performed by special inspectors qualified to inspect the work to be inspected. The qualifications of the special inspector shall be acceptable to the commissioner.

1705.2 Steel construction. The special inspections of new installations and alterations to existing installations of steel elements in buildings and structures shall be as required by this section and Table 1705.2. Where required, special inspection of steel shall also comply with Section 1711.

Exceptions:

1. Special inspection of the steel fabrication process shall not be required where the fabricator does not perform any welding, thermal cutting or heating operation of any kind as part of the fabrication process. In such cases, the fabricator shall be required to submit a detailed procedure for material control that demonstrates the fabricator's ability to maintain suitable records and procedures such that, at any time during the fabrication process, the material specification, grade and mill test reports for the main stress-carrying elements are capable of being determined.
2. When specified by the registered design professional, the special inspector shall have the option to perform periodic inspections during welding of the following items, provided the materials, welding procedures and qualifications of welders are verified prior to the start of the work; periodic inspections are made of the work in progress and a visual inspection of all welds is made prior to completion or prior to shipment of shop welding.
 - 2.1. Single-pass fillet welds not exceeding $\frac{5}{16}$ inch (7.9 mm) in size.
 - 2.2. Floor and roof deck welding.
 - 2.3. Welded studs when used for structural diaphragm.
 - 2.4. Welded sheet steel for cold-formed steel members.
 - 2.5. Welding of stairs and railing systems.
 - 2.6. Welding of connections where the calculated connection strength is at least twice the required strength of the connection. Such connections shall be specifically indicated on the approved construction documents.

**TABLE 1705.2
§§§ REQUIRED SPECIAL INSPECTION OF STEEL CONSTRUCTION**

TYPE	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION	REFERENCED STANDARD ^a	BC REFERENCE
1. Material verification of high-strength bolts, nuts and washers:				
a. Identification markings to conform to ASTM standards specified in the approved construction documents.	—	X	AISC 360, Section A3.3; applicable ASTM material specifications; and RCSC Specification for Structural Joints Using High-Strength Bolts Section 2	—

SPECIAL INSPECTIONS AND TESTS

b. Manufacturer's certificate of compliance required.	—	X	RCSC Specification for Structural Joints Using High-Strength Bolts Section 2.1	—
2. Inspection of high-strength bolting:				
a. Snug-tight joints.	—	X	AISC 360 Section M2.5; and RCSC Specification for Structural Joints Using High-Strength Bolts Section 9	1705.2.3
b. Pre-tensioned and slip-critical joints using turn-of-nut with matchmarking, twist-off bolt or direct tension indicator methods of installation.	—	X		
c. Pre-tensioned and slip-critical joints using turn-of-nut without matchmarking or calibrated wrench methods of installation.	X	—		
d. Pre-installation verification testing.	X	—	Specification for Structural Joints Using High-Strength Bolts Section 8.2	1705.2.3.1
3. Material verification of structural steel and cold formed steel deck:				
a. For structural steel, identification markings to conform to AISC 360.	—	X	AISC 360 Sections A3.1, N2.1, N3.2 (a) and (k)(1)	—
b. For other steel, identification markings to conform to ASTM standards specified in the approved construction documents.	—	—	Applicable ASTM Standards	—
c. Manufacturers' certified mill test reports.	—	X	Applicable ASTM material standards	—
4. Material verification of weld filler materials:				
a. Identification markings to conform to AWS specification in the approved construction documents.	—	—	AISC 360 Section A3.5 and N3.2(e), and applicable AWS A5 documents; and AWS D1.1 5.3.1 and approved contract documents	—
b. Manufacturer's certificate of compliance required.	—	—	AISC 360 Section A3.5	—
5. Inspection of welding:				
a. Structural steel:	—	—	—	—
1) Complete and partial penetration groove welds.	X	—	AWS D1.1	1705.2.1
2) Multipass fillet welds.	X	—		
3) Single-pass fillet welds $> \frac{5}{16}$ ".	X	—		
4) Plug and slot welds.	X	—		
5) Single-pass fillet welds $\leq \frac{5}{16}$ ".	—	X		
6) Floor and roof deck welds.	—	X	AWS D1.3	—
7) Cold-formed steel welds.	—	X	AWS D1.3	—
b. Reinforcing steel:	—	—	AWS D1.4	1903.6.2
1) Pre-welding verification of base metal.	—	X		

2)	Reinforcing steel-resisting flexural and axial forces in intermediate and special moment frames, and boundary elements of special reinforced concrete shear walls and shear reinforcement.	X	—
3)	Shear reinforcement.	X Note a	—
4)	Other reinforcing steel.	—	X Note b

(continued)

TABLE 1705.2—continued
REQUIRED SPECIAL INSPECTION OF STEEL CONSTRUCTION

TYPE	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION	REFERENCED STANDARD ^a	BC REFERENCE
6. Inspection of steel frame joint details for compliance with approved construction documents:				
a. Details such as bracing and stiffening.	—	X	—	1705.2.2
b. Member locations.	—	X		
c. Application of joint details at each connection.	—	X		

For SI: 1 inch = 25.4 mm.

- a. A minimum of 10% of shear studs shall be verified for strength of welded connection. If failure is evident on one or more, then the strength of all shear studs shall be verified.
- b. Welding of indirect and direct butt joints shall be continuously inspected.

1705.2.1 Welding. Welding inspection and welding inspector qualification shall be in compliance with this section and the rules of the department.

1705.2.1.1 Structural steel. Welding inspection and welding inspector qualification for structural steel shall be in accordance with AWS D1.1.

1705.2.1.2 Cold-formed steel deck. Welding inspection and welding inspector qualification for cold-formed steel floor and roof decks shall be in accordance with SDI QA/QC.

1705.2.1.3 Reinforcing steel. Welding inspection and welding inspector qualification for reinforcing steel shall be in accordance with AWS D1.4.

1705.2.2 Details. The special inspector shall perform an inspection of the steel frame to verify compliance with the details shown on the approved construction documents, such as bracing, stiffening, member locations and proper application of joint details at each connection, including connections designed for seismic effects.

1705.2.3 High-strength bolts. Installation of high-strength bolts shall be inspected in accordance with this section and AISC 360.

Exception: Installation of high strength bolts in bearing type connections where the calculated connection strength is at least twice the required strength of the connection. Inspection of such connections shall be made under Section 1705.2.2. Such connections shall be clearly indicated on the approved construction documents.

1705.2.3.1 General. While the work is in progress, the special inspector shall determine that the requirements for bolts, nuts, washers and paint; bolted parts and installation and tightening in such standards are met. For bolts requiring pretensioning, the special inspector shall observe the preinstallation testing and calibration procedures when such procedures are required by the installation method or by project plans or specifications; determine that all plies of connected materials have been drawn together and properly snugged and monitor the installation of bolts to verify that the selected procedure for installation is properly used to tighten bolts. For joints required to be tightened only to the snug-tight condition, the special inspector need only verify that the connected materials have been drawn together and properly snugged.

1705.2.3.2 Periodic inspection. Inspection of bolt installation for pretensioning is permitted to be performed on a periodic basis when using the turn-of-nut method with matchmarking techniques, the direct tension indicator method or the alternate design fastener (twist-off bolt) method. Joints designated as snug tight need be inspected only on a periodic basis.

1705.2.3.3 Continuous inspection. Inspection of bolt installation for pretensioning using the calibrated wrench method or the turn-of-nut method without matchmarking shall be performed on a continuous basis.

††† **1705.2.4 Special inspection for seismic resistance.** In addition to the other special inspection requirements of this code, special inspections of structural steel seismic force resisting systems and structural steel elements shall be performed in accordance with Sections 1705.2.4.1 and 1705.2.4.2.

1705.2.4.1 Seismic force-resisting systems. Special inspections of structural steel in the seismic force-resisting systems of buildings and structures assigned to Seismic Design Category B, C or D shall be performed in accordance with the quality assurance requirements of AISC 341.

Exception: Special inspections per Section 1705.2.4.1 are not required in the seismic force-resisting systems of buildings and structures assigned to Seismic Design Category B or C that are not specifically detailed for seismic resistance, with a response modification coefficient, R , of 3 or less, excluding cantilever column systems.

1705.2.4.2 Structural steel elements. Special inspections of structural steel elements in the seismic force-resisting systems of buildings and structures assigned to Seismic Design Category B, C or D other than those covered in Section 1705.2.4.1 of this code, including struts, collectors, chords and foundation elements, shall be performed in accordance with the quality assurance requirements of AISC 341.

Exception: Special inspections of structural steel elements per Section 1705.2.4.2 are not required in the seismic force-resisting systems of buildings and structures assigned to Seismic Design Category B or C with a response modification coefficient, R , of 3 or less.

1705.2.5 Testing for seismic resistance of structural steel. Nondestructive testing for seismic resistance shall be in accordance with Section 1705.2.5.1 or 1705.2.5.2, as applicable.

1705.2.5.1 Seismic force-resisting systems. Nondestructive testing of structural steel in the seismic force-resisting systems of buildings and structures assigned to Seismic Design Category B, C, or D shall be performed in accordance with the quality assurance requirements of AISC 341.

Exception: Nondestructive testing is not required in the seismic force-resisting systems of buildings and structures assigned to Seismic Design Category B or C that are not specifically detailed for seismic resistance, with a response modification coefficient, R , of 3 or less, excluding cantilever column systems.

1705.2.5.2 Structural steel elements. Nondestructive testing of structural steel elements in the seismic force-resisting systems of buildings and structures assigned to Seismic Design Category B, C or D other than those covered in Section 1705.2.5.1, including struts, collectors, chords and foundation elements, shall be performed in accordance with the quality assurance requirements of AISC 341.

Exception: Nondestructive testing of structural steel elements is not required in the seismic force-resisting systems of buildings and structures assigned to Seismic Design Category B or C with a response modification coefficient, R , of 3 or less.

1705.2.6 Cold-formed steel construction. Special inspections for prefabricated and site built cold-formed steel construction and assemblies used structurally shall be as required by this section and Table 1705.2.6.

††† ***TABLE 1705.2.6**
REQUIRED SPECIAL INSPECTION OF COLD-FORMED STEEL CONSTRUCTION

SPECIAL INSPECTIONS AND TESTS

TYPE	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION	REFERENCED STANDARD	BC REFERENCE
1. Material Verification:				
a. Verify that identification markings conform to AISI S240 and as specified in the approved construction documents.	X		AISI S240, Section D6.5	
b. Verify that material is clean, straight and undamaged.		X		
2. Inspection of general framing:				
a. Verify that member sizes conform to the approved construction documents.		X	AISI S240 Section C	
b. Verify that member layout conforms to the approved construction documents.		X	AISI S240 Section C	
c. Verify that proper bearing lengths are provided in accordance with approved construction documents.		X	AISI S240 Section C	
d. Verify that punched holes and sheared or flame cut edges of material in members are clean and free from notches and burred edges.		X	AISI S240 Section C	
3. Inspection of framing connections and anchorages:				
a. Verify that screws, bolts, and other fasteners conform to approved construction document requirements for diameter, length, quantity, spacing, edge distance, and location.		X	AISI S240, Section D6.7	
b. Verify that manufactured connectors, such as joist hangers, caps, straps, clips, ties, hold-downs, and anchors conform to approved construction document requirements for manufacturer, type, gauge, and fastener requirements.		X	AISI S240 Section B1.5 and Section C4	
c. Post-installed connections to concrete.	X		AISI S240 Section D6.9	
4. Inspection of welding:				
a. Inspect welds in accordance with S240 Section D6.6.		X	AWS D1.3, AISI S240 Section D6.6	
b. Additional requirements for welds performed as a part of a lateral force-resisting system.	X Note a		AISI S240 Section D6.9	
5. Bracing:				
a. Verify that temporary bracing, shoring, jacks, etc., are installed, modified and not removed until no longer necessary, in accordance with the approved construction documents and approved erection drawings as required by Sections 3305.8.6.8 and 3305.8.7.5.		X	AISI S240 Section E6	

SPECIAL INSPECTIONS AND TESTS

b. Verify that permanent bracing, web stiffeners, bridging, blocking, wind bracing, etc., are installed in accordance with the approved construction documents and approved erection drawings, as required by Sections 3305.8.6.8 and 3305.8.7.5.		X	AISI S240 Section E6	
---	--	---	-------------------------	--

(continued)

††† *REQUIRED SPECIAL INSPECTION OF COLD-FORMED STEEL CONSTRUCTION

TYPE	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION	REFERENCED STANDARD	BC REFERENCE
5. Bracing: (continued)				
c. Where a cold-formed steel truss clear span is 60 feet (18 288 mm) or greater, the special inspector shall verify that the temporary installation restraint/bracing and the permanent individual truss member restraint/bracing are installed in accordance with the approved truss submittal package.		X	AISI S240 Section E6	2211.1.3.2
6. Pre-installation Document Submittals		X	AISI S240, Section D3	
7. Lateral Force-Resisting System Additional Requirements		X	AISI S240 Section D6.9	

Note a: In accordance with AISI S240 Section D6.9.1, continuous special inspection of weld fit-up in lateral force-resisting systems may be reduced to periodic special inspection upon fulfillment of the conditions of Section D6.9.1.

*Table 1705.2.6 was amended by [Local Law 148 of 2021](#). This law has an effective date of November 7, 2022.

1705.2.7 Open-web steel joists and joist girders. Special inspections of open web steel joists and joist girders in buildings, structures, and portions thereof shall be in accordance with Table 1705.2.7.

**TABLE 1705.2.7
REQUIRED SPECIAL INSPECTIONS OF OPEN-WEB STEEL JOISTS AND JOIST GIRDERS**

TYPE	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION	REFERENCED STANDARD	BC REFERENCE
1. Inspection of member sizes and locations.	—	X		
2. Inspection of end connections – welded or bolted	—	X	SJI Specifications	2207.1
3. Inspection of standard bridging, horizontal or diagonal	—	X	SJI Specifications	2207.1
4. Inspection of bridging, horizontal or diagonal, that differs from the SJI specifications listed in Section 2207.1	—	X	SJI Specifications	

††† **1705.3 Concrete construction.** The special inspections and verifications for concrete construction shall be as required by this section and Table 1705.3.

Exceptions:

1. Special inspection shall not be required when specifically indicated as not required on the approved construction documents for:

SPECIAL INSPECTIONS AND TESTS

- 1.1. Nonstructural concrete slabs supported directly on the ground, including prestressed slabs on grade, where the effective prestress in the concrete is less than 150 psi (1.03 Mpa).
- 1.2. Concrete foundation(s) for lightweight fences and recreational equipment.
- 1.3. Concrete patios, site furnishings, garden walls, driveways, sidewalks and similar construction.
2. Testing required by Table 1705.3, Item 6, may be waived by the registered design professional who prepared the approved structural construction documents when such waiver is specifically indicated on such construction documents in the following cases:
 - 2.1. Where the total concrete placement on a given project is less than 50 cubic yards (38.2 m³).
 - 2.2. Isolated spread concrete footings of R-3 buildings three stories or less above grade plan that are fully supported on earth or rock.
 - 2.3. Continuous concrete footings supporting walls of R-3 buildings three stories or less above grade plan that are fully supported on earth or rock where the structural design of the footing is based on a specified compressive strength, f'_c , no greater than 2,500 pounds per square inch (psi) (17.2 Mpa), and the compressive strength used in the footing construction is at least 4,000 psi (27.6 Mpa).

TABLE 1705.3
REQUIRED SPECIAL INSPECTIONS AND TESTS OF CONCRETE CONSTRUCTION

TYPE	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION	REFERENCED STANDARD	BC REFERENCE	CONCRETE SPECIAL INSPECTOR (CAST-IN-PLACE, PRECAST, & PRESTRESSED)	LICENSED CONCRETE TESTING LABORATORY
1. Inspect reinforcement, including prestressing tendons and verify placement.	—	X	ACI 318: 3.5, 7.1 – 7.7	1903.6, 1907.1, 1907.7, 1911.4	X	
2. Inspection of reinforcing steel welding in accordance with Table 1705.2, Item 5b.	—	—	AWS D1.4 ACI 318: 3.5.2	1903.6.2		
3. Inspect anchors cast in concrete.	X	—	ACI 318: 17.8.2	1901.3	X	
4. Inspect anchors post-installed in hardened concrete members ^c . a. Adhesive anchors installed in horizontally or upwardly inclined orientations to resist sustained tension loads. b. Mechanical anchors and adhesive anchors not defined in 4a.	X —	— X	ACI 318: 17.8		X	
5. Verifying use of required design mix.	—	—	ACI 318: Ch. 4, 5.2–5.4	1904, 1905.2–1905.4, 1911.3		X
6. Prior to concrete placement, fabricate specimens for strength tests, perform slump and air content tests, and determine the temperature of the concrete. Determine water content when required.	—	—	ASTM C 172 ASTM C 31 ACI 318: 5.6, 5.8 (Notes a, b)	1905.6, 1911.10		X
7. a. Inspect concrete and shotcrete placement for proper application techniques.	X	—	ACI 318: 5.9, 5.10	1905.9, 1905.10, 1911.6,	X	

SPECIAL INSPECTIONS AND TESTS

b. For concrete pumped through a placement boom: Following the lubrication of the concrete placement boom and prior to Contractor's commencement of the concrete pour, observe and document as part of the special inspection of the concrete placement whether the material exiting the hose is concrete exhibiting a uniform matrix of aggregate.				1911.7, 1911.8		
8. Verify maintenance of specified curing temperature and techniques. Monitoring of in-place temperatures per thermal protection plan when required.	—	X	ACI 318: 5.11–5.13	1905.11, 1905.13, 1911.9	X	
9. Inspect prestressed concrete for: A. Application of prestressing forces; and B. Grouting of bonded prestressing tendons.	X X	—	ACI 318: 18.20 ACI 318: 18.18.4	—	X	
10. Inspect erection of precast concrete members.	—	X	ACI 318: Ch. 16	—	X	
11. Verify in-situ concrete strength, prior to stressing of tendons in post-tensioned concrete and prior to removal of shores and forms from beams and structural slabs.	—	X	ACI 318: 6.2	1906.2	—	X
12. Inspect formwork for shape, location, and dimensions of the concrete member being formed.	—	X	ACI 318: 6.1.1	1906.2	X	

a. Standard sampling rate shall be in accordance with Section 1905.6.2.

b. Four-inch by 8-inch cylinders may be accepted in lieu of 6-inch by 12-inch cylinders at the option of the engineer of record.

c. Specific requirements for special inspection shall be included in the research report for the anchor issued by an approved source in accordance with 17.8.2 in ACI 318, or other qualification procedures. Where specific requirements are not provided, special inspection requirements shall be specified by the registered design professional and shall be approved by the commissioner prior to the commencement of the work.

1705.3.1 Materials. In the absence of sufficient data or documentation providing evidence of conformance to quality standards for materials in Chapter 3 of ACI 318, the commissioner shall require testing of materials in accordance with the appropriate standards and criteria for the material in Chapters 19 and 20 of ACI 318. Weldability of reinforcement, except that which conforms to ASTM A 706, shall be determined in accordance with the requirements of Section 1903.6.2 of this code.

1705.3.2 Concrete construction. When the specified compressive strength of concrete in a column is greater than 1.4 times that specified for a floor system, the special inspections for concrete construction shall also comply with the requirements of Section 1908.1.4.

1705.4 Masonry construction. Masonry construction shall be inspected and verified in accordance with the requirements of Sections 1705.4.1 through 1705.4.3, depending on the risk category of the building or structure.

Exception: Special inspections shall not be required for:

1. Masonry fireplaces, masonry heaters or masonry chimneys installed or constructed in accordance with Section 2111, 2112, or 2113, respectively.
2. Alterations to existing masonry, where the quantity of replaced masonry is less than 10 square feet (0.93 m²) in any 100 square feet (9.3 m²) of wall area, when specifically exempted on the approved construction documents.

1705.4.1 Empirically designed masonry, glass unit masonry and masonry veneer in buildings in Risk Category IV. The minimum inspection program for empirically designed masonry, glass unit masonry and masonry

SPECIAL INSPECTIONS AND TESTS

veneer designed in accordance with Section 2109, 2110, or Chapter 14 of this code, respectively, or by Appendix A, Chapter 13 or 12[‡] of TMS 402 as modified by Chapter 21 of this code, in structures classified as Risk Category IV, in accordance with Table 1604.5, shall comply with Table 1705.4.1 of this code.

**TABLE 1705.4.1
LEVEL 2[‡] REQUIRED SPECIAL INSPECTIONS AND TESTS OF MASONRY CONSTRUCTION**

MINIMUM TESTS					
Verification of Slump flow and Visual Stability Index (VSI) as delivered to the project site in accordance with TMS 602 Article 1.5.B.1.b.3 for self-Consolidating grout.					
Verification of $f'm$ and f'_{AAC} in accordance with TMS 602 Article 1.4 B prior to construction, except where specifically exempted by this Code.					
TYPE	FREQUENCY		REFERENCE FOR CRITERIA		
	Continuous Special Inspection	Periodic Special Inspection	BC Section	TMS 402	TMS 602
1. Verify compliance with required inspection provisions of the construction documents and the approved submittals.	—	X	—	—	Art. 1.5
2. As masonry construction begins, the following shall be verified to ensure compliance:					
a. Proportions of site-prepared mortar.	—	X	—	—	Art. 2.6A
b. Construction of mortar joints.	—	X	—	—	Art. 3.3B
c. Grade and size of prestressing tendons and anchorages.	—	X	—	—	Art. 2.4B, 2.4H
d. Location of reinforcement, connectors, prestressing tendons and anchorages.	—	X	—	—	Art. 3.4, 3.6A
e. Prestressing technique.	—	X	—	—	Art. 3.6B
3. Prior to grouting, the following shall be verified to ensure compliance:					
a. Grout space.	—	X	—	—	Art. 3.2D
b. Grade, type, and size of reinforcement and anchor bolts, and prestressing tendons and anchorages.	—	X	—	—	—
c. Placement of reinforcement, connectors, and prestressing tendons and anchorages.	—	X	—	Sec. 1.13	Art. 3.4
d. Proportions of site-prepared grout and prestressing grout for bonded tendons.	—	X	—	—	Art. 2.6B
e. Construction of mortar joints.	—	X	—	—	Art. 3.3B
4. Verify during construction:					
a. Size and location of structural elements.	—	X	—	—	Art. 3.3F
b. Type, size and location of anchors, including other details of anchorage of masonry to structural members, frames or other construction.	—	X	—	Sec. 1.2.1(e), 14.4.3	—

SPECIAL INSPECTIONS AND TESTS

c. Welding of reinforcement.	X	—	—	Sec. 1.2.1(g), 6.1.6.1.2, 6.1.7.3, 13.7	—
d. Preparation, construction and protection of masonry during cold weather (temperature below 40°F) or hot weather (temperature above 90°F).	—	X	Sec. 2104.2, 2104.3	Sec 3.1.3	Art. 1.8C, 1.8D
e. Preparation of any required grout specimens, mortar specimens and/or prisms shall be observed.	—	X	—	Art. 3.1.3	Art. 1.4
f. Application and measurement of prestressing force.	X	—	—	—	Art. 3.6B
g. Placement of grout and prestressing grout for bonded tendons is in compliance.	X	—	—	—	Art. 3.5, 3.6 C

(continued)

TABLE 1705.4.1—continued
LEVEL 2^a REQUIRED SPECIAL INSPECTIONS AND TESTS OF MASONRY CONSTRUCTION

MINIMUM TESTS					
Verification of Slump flow and Visual Stability Index (VSI) as delivered to the project site in accordance with TMS 602 Article 1.5.B.1.b.3 for self-Consolidating grout.					
Verification of $f'm$ and f'_{AAC} in accordance with TMS 602 Article 1.4 B prior to construction, except where specifically exempted by this Code.					
TYPE	FREQUENCY		REFERENCE FOR CRITERIA		
	Continuous Special Inspection	Periodic Special Inspection	BC Section	TMS 402	TMS 602
h. Placement of AAC masonry units and construction of thin-bed mortar joints.	X ^b	X ^c	—	—	Art. 3.3 B.9, 3.3 F.1.b
i. Grout space is clean.	—	X	—	—	Art. 3.2D
j. Placement of reinforcement and connectors, and prestressing tendons and anchorages.	—	X	—	Sec. 6.1.3 and 10.8	Art. 3.4
k. Proportions of site-prepared grout and prestressing grout for bonded tendons.	—	X	—	—	Art. 2.6B
l. Construction of mortar joints.	—	X	—	—	Art. 3.3B
5. Observe preparation of grout specimen, mortar specimen, and/or prisms.	—	X	—	—	Art: 1.4B2a3, 1.4B2b3, 1.4B2c3, 1.4B3, 1.4B4, 2.1A

For SI: °C = [(°F) - 32]/1.8.

a. The specific standards referenced are those listed in Chapter 35.

b. Required for the first 5,000 square feet of AAC masonry.

c. Required after the first 5,000 square feet of AAC masonry.

1705.4.2 Masonry in buildings three stories or less in height in Risk Categories I, II, or III. The minimum special inspection program for structures classified as Risk Categories I, II, or III, in accordance with Table 1604.5, which are three stories or less in height, shall comply with Table 1705.4.1.

1705.4.3 Masonry in buildings greater than three stories in height, and engineered masonry in Risk Category IV. The minimum special inspection program for masonry designed in accordance with Section 2107 or 2108, or by chapters other than Chapters 12 or 13, or Appendix A[‡] of TMS 402 as modified by Chapter 21 of this code, in structures classified as Risk Category IV, in accordance with Table 1604.5 of this code, and masonry in buildings greater than three stories in height, shall comply with Table 1705.4.3 of this code.

Exception: Alterations to existing masonry walls where the quantity of masonry involved in any one given location does not exceed 50 contiguous square feet (4.64 m²) shall be permitted to be subject to level 2[‡] special inspection in accordance with Table 1705.4.1 when specifically identified as such on the approved construction documents by the registered design professional of record.

+++**TABLE 1705.4.3**
LEVEL 3[‡] REQUIRED SPECIAL INSPECTIONS AND TESTS OF MASONRY CONSTRUCTION

MINIMUM TESTS					
Verification of $f'm$ and f'_{AAC} in accordance with TMS 602 Article 1.4 B prior to construction and every 5,000 sq. ft. during construction.					
Verification of proportions of materials in premixed or preblended mortar, prestressing grout, and grout other than self-consolidating grout, as delivered to the project site in accordance with TMS 602 Article 1.5B.					
Verification of Slump flow and Visual Stability Index (VSI) as delivered to the project site in accordance with TMS 602 Article 1.5.B.1.b.3 for self-consolidating grout.					
MINIMUM SPECIAL INSPECTION					
TYPE	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION	REFERENCE FOR CRITERIA		
			BC Section	TMS 402	TMS 602 ^a
1. Verify compliance with required inspection provisions of the construction documents and the approved submittals.	—	X	—	—	Art. 1.5
2. Verify that the following are in compliance:					
a. Proportions of site-prepared mortar, grout and prestressing grout for bonded tendons.	—	X	—	—	Art. 2.6 A
b. Grade, type, and size of reinforcement and anchor bolts, and prestressing grout for bonded tendons.	—	X [‡]	—	Sec. 6.1.2, 10.8	Art. 2.4, 3.4
c. Placement of masonry units and construction of mortar joints.	— [‡]	X	—	—	Art. 3.3 B
d. Placement of reinforcement, connectors and prestressing tendons and anchorages.	—	X	—	Sec. 6.1.2, 10.8	Art. 3.4, 3.6 A
e. Grout space prior to grouting.	X	—	—	—	Art. 3.2 D
f. Placement of grout and prestressing grout for bonded tendons.	X	—	—	—	Art. 3.5
g. Placement of prestressing grout.	X	—	—	—	Art. 3.6 C
h. Size and location of structural elements.	—	X	—	—	Art. 3.3 F

SPECIAL INSPECTIONS AND TESTS

i. Type, size and location of anchors, including other details of anchorage of masonry to structural members, frames or other construction.	X	—	—	Sec. 1.2.1(e), 14.4.3	—
j. Welding of reinforcement.	X	—	—	Sec. 1.2.1(g), 6.1.6.1.2, 6.1.7.3, 13.7	—
k. Preparation, construction and protection of masonry during cold weather (temperature below 40°F) or hot weather (temperature above 90°F).	—	X	Sec. 2104.2, 2104.3	—	Art. 1.8 C, 1.8 D
l. Application and measurement of prestressing force.	X	—	—	—	Art. 3.6 B
m. Placement of AAC masonry units and construction of thin-bed mortar joints.	X	—	—	—	Art. 3.3 B.9, 3.3 F.1.b
n. Properties of thin-bed mortar for AAC masonry.	X	—	—	—	Art. 2.1 C.1
6. Observe preparation of grout specimens, mortar specimens, and/or prisms.	X	—	—	Sec. 3.1.3	Art. 1.4

For SI: °C = [(°F) - 32]/1.8, 1 square foot = 0.0929 m².

a. The specific standards referenced are those listed in Chapter 35.

††† **1705.5 Wood construction.** Special inspections of site-built assemblies shall be in accordance with Sections 1705.5.1 through 1705.5.4. Special inspections of the fabrication process of prefabricated wood structural elements and assemblies shall be in accordance with Section 1704.2.

1705.5.1 High-load diaphragms. High-load diaphragms designed in accordance with Table 2306.2(2) shall be subject to special inspections as indicated in Section 1704.1. The special inspector shall inspect the wood structural panel sheathing to ascertain whether it is of the grade and thickness shown on the approved construction documents. Additionally, the special inspector must verify the nominal size of framing members at adjoining panel edges, the nail or staple diameter and length, the number of fastener lines and that the spacing between fasteners in each line and at edge margins agrees with the approved construction documents.

1705.5.2 Metal-plate-connected wood trusses. In addition to the requirements of Section 1704.1, metal-plate-connected wood trusses shall be subject to special inspection in accordance with Sections 1705.5.2.1 and 1705.5.2.2.

††† **1705.5.2.1 Erection.** The use of all metal-plate-connected wood trusses shall be subject to special inspection for compliance with the approved construction documents, the requirements of Sections 1704.1, and the following:

1. All installed materials shall be clean, straight and otherwise undamaged. Members and parts shall not be stretched, bent, or otherwise distorted unless such forming is in the integral part of the design. The special inspector shall ensure that damaged members are not used for construction.
2. Profiles of members used structurally shall conform to the dimensions specified in the approved construction documents. The installation shall be inspected for compliance with the approved construction documents regarding locations, positions, beam separators, bearing surfaces, fasteners, screws, bolts and bracing, as applicable.
3. Temporary bracing, shoring, jacks, etc., shall not be removed until the special inspector determines that the construction conforms with the approved construction document.
4. Where prefabricated metal-plate-connected wood trusses are utilized, such prefabricated wood structural elements and assemblies shall also comply with Section 1704.2. Where any metal-plate connectors are

utilized in site-built assemblies, such connections and assemblies shall be subject to special inspection for compliance with the requirements of the approved construction documents and manufacturers' instructions.

1705.5.2.2 Metal-plate-connected wood trusses spanning 60 feet or greater. Where a truss clear span is 60 feet (18 288 mm) or greater, the special inspector shall verify that the temporary installation restraint/bracing and the permanent individual truss member restraint/bracing are installed in accordance with the approved truss submittal package.

††† **1705.5.3 Prefabricated wood I-joists and/or glued prefabricated parallel chord wood trusses.** The fabrication of prefabricated wood I-joists and glued prefabricated parallel chord wood trusses shall be subject to special inspections in accordance with Section 1704.2 and the requirements of Section 2303.1.2.

††† **1705.5.3.1 Erection.** The erection of prefabricated wood I-joists and/or glued prefabricated parallel chord wood trusses shall be subject to special inspection for compliance with the approved construction documents, the requirements of Sections 1704.1, and the following:

1. All installed materials shall be clean, straight and otherwise undamaged. Members and parts shall not be stretched, bent, or otherwise distorted unless such forming is in the integral part of the design. The special inspector shall ensure that damaged members are not used for construction.
2. Profiles of members used structurally shall conform to the dimensions specified in the approved construction documents. The installation shall be inspected for compliance with the approved construction documents regarding locations, positions, beam separators, bearing surfaces, fasteners, screws, bolts and bracing, as applicable.
3. The size, location, and number of penetrations shall be inspected for compliance with the approved construction documents and manufacturers' instructions.
4. Temporary bracing, shoring, jacks, etc., shall not be removed until the special inspector determines that they are no longer needed.

1705.5.4 Other structural wood construction. All other structural wood construction shall be subject to progress inspections for the structural frame to the extent required by Section 110.3.3.

1705.5.5 Special inspection for seismic resistance. For the seismic force-resisting systems of structures assigned to Seismic Design Category C or D:

1. Continuous special inspection shall be required during field gluing operations of elements of the seismic force resisting system.
2. Periodic special inspection shall be required for nailing, bolting, anchoring and of other elements of the seismic force-resisting system, including wood shear walls, wood diaphragms, drag struts, braces, shear panels and hold-downs.

Exception:

1. Special inspections are not required for wood shear walls, shear panels and diaphragms, including nailing, bolting, anchoring and other fastening to other elements of the seismic force-resisting system, where the fastener spacing of the sheathing is more than 4 inches (101.6 mm) on center.
2. Where adhesive anchors require continuous special inspection elsewhere in this code, continuous special inspection shall be provided.

1705.5.6 Type IV construction. Special inspections of Type IV construction utilizing cross-laminated timber or structural composite lumber elements shall be in accordance with Table 1705.5.6.

††† **TABLE 1705.5.6**
REQUIRED SPECIAL INSPECTIONS OF TYPE IV CONSTRUCTION
UTILIZING CROSS-LAMINATED TIMBER OR STRUCTURAL COMPOSITE LUMBER

TYPE	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION	REFERENCED STANDARD	BC† CODE REFERENCE
------	-------------------------------------	-----------------------------------	------------------------	-----------------------

SPECIAL INSPECTIONS AND TESTS

1. Inspection of anchorage and connections of mass timber construction to timber deep foundation systems	—	X		1705.7, 2308.3, 2304.10
2. Inspect erection of mass timber, including material verification	—	X	PRG-320, ASTM D5456	2303.1.4, 2303.1.10
3. Inspection of connections where installation methods are required to meet design loads			NDS 11-14 ANSI/ASME Standard B18.2.1, B18.6.1 As per connection manufacturer's installation instructions	2304.10, 2304.11.1.1 to 2304.11.4
3.1. Threaded fasteners	—	X		
3.1.1. Verify use of proper installation equipment	—	X		
3.1.2. Verify use of pre-drilled holes where required	—	X		
3.1.3. Inspect screws, including diameter, length, head type, spacing, installation angle, and depth	—	X		
3.2. Adhesive anchors installed in horizontal or upwardly inclined orientation to resist sustained tension loads	X	—		
3.3. Adhesive anchors not defined in 3.2	—	X		
3.4. Bolted connections	—	X		
3.5. Concealed connections	—	X		
3.6. Other fastener types	—	X		
4. Inspection of connections where installation methods are required to meet the fire resistance design in Section‡ 2304.10.8	—	X	NDS 16.3	

1705.6 Subsurface conditions. The special inspections for existing subsurface conditions, fill placement, load-bearing requirements, subsurface investigations, borings and test pits shall be performed in accordance with this section and Table 1705.6 of this code. The approved geotechnical report and construction documents, as applicable, shall be used to determine compliance. Special inspection reports of borings and/or test pits shall be in accordance with ASTM D 1586, ASTM D 1587 and Section 1803.5 of this code.

**TABLE 1705.6
REQUIRED SPECIAL INSPECTIONS OF SUBSURFACE CONDITIONS**

TYPE	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION
1. Special inspection of subsurface investigations, borings and test pits. Boring and test pit operations, and alternative investigation methods, shall be subject to continuous special inspection to verify compliance with Section 1803. Soil sample recovery operations for test pits shall be subject to continuous special inspection to verify compliance with Section 1803. Exceptions: - Cone penetrometer testing used as an alternative investigation method per Section 1803.5.2 shall be subject to periodic special inspection. - Existing boring, test pit and alternative investigation records that have been deemed acceptable to the commissioner in accordance with Section 1803.4.2 are not subject to special inspection.	X	—
2. During fill placement. During placement and compaction of the fill material, the special inspector shall determine that the material being used and the maximum lift thickness comply with the approved geotechnical report, as specified in Section 1804.6‡.	X	—

SPECIAL INSPECTIONS AND TESTS

3. Evaluation of in-place density. The special inspector shall determine that the in-place dry density of the compacted fill complies with the approved construction documents.	X	—
4. Subgrade inspection. Immediately prior to placement of each and every footing, foundation, fill or other supporting materials, the special inspector shall determine that the site has been prepared and is in accordance with the approved geotechnical report.	—	X

SPECIAL INSPECTIONS AND TESTS

1705.7 Deep foundation elements. Installation and testing of deep foundation elements shall be subject to special inspections as specified in Table 1705.7. The approved geotechnical report and construction documents prepared by the registered design professionals shall be used to determine compliance. A special inspector shall be present during the installation and testing of each deep foundation element. The frequency shall be in accordance with the frequency specified in Tables 1705.7, 1705.7.1, 1705.7.2 and 1705.7.3, as applicable. Submittal of records of special inspections required by this section shall be in accordance with Section 1705.7.5.

**TABLE 1705.7
REQUIRED SPECIAL INSPECTIONS AND TESTS OF DEEP FOUNDATION ELEMENTS**

INSTALLATION	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION
1. Verify the identifying designation of the deep foundation element and record the date of installation, including the start and end times.	X	—
2. Verify the size, material, and allowable capacity as specified in the construction documents.	X	—
3. Record the elevation of the minimum required depth of penetration and record the final tip elevation and butt elevation.	X	—
4. Record the elevation of splices and note whether or not the splices were installed and located in compliance with the construction documents.	X	—
LOAD TEST		
5. For load test requiring a load or reaction frame, inspect the construction of the load or reaction frame. Record the results of the inspection and note whether or not the frame complies with the construction documents prepared by the registered design professional.	—	X
6. Record the identifying designation for the element being tested, and the date of the testing, including the start and end time.	X	—
7. Record the method of performing the test, including the equipment being used, as well as the test results, noting whether or not the method of testing and the test results comply with the requirements of Sections 1810, 1811, and 1812 and the construction documents.	X	—

1705.7.1 Driven and vibrated deep foundation elements. The installation and testing of driven and vibrated deep foundation elements shall be subject to special inspections as specified in Table 1705.7 and Table 1705.7.1.

**TABLE 1705.7.1
REQUIRED SPECIAL INSPECTIONS OF DRIVEN AND VIBRATED DEEP FOUNDATION ELEMENTS**

INSTALLATION	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION
1. For driven deep foundation elements, record the type and size of hammer and record the number of blows per foot of penetration from the start of driving until the final blow count required by the construction documents is reached.	X	—
2. For vibrated deep foundation elements, record the equipment and method used for installation and record the time for each foot of penetration from the start of installation until completion.	X	—

1705.7.2 Drilled deep foundation elements. The installation and testing of drilled deep foundation elements shall be subject to special inspections as specified in Table 1705.7 and Table 1705.7.2.

**TABLE 1705.7.2
REQUIRED SPECIAL INSPECTIONS OF DRILLED DEEP FOUNDATION ELEMENTS**

INSTALLATION	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION
1. Record the equipment and method used for installation and record the time for each foot of penetration from the start of installation until completion.	X	—
2. For deep foundation elements requiring a rock socket: 2.1. Record the equipment and method of drilling the rock socket. Record the top elevation and bottom elevation of rock socket, as well as the time for each foot of drilling the rock socket. 2.2. Inspect the rock socket to verify rock quality as required by Section 1812 and the construction documents. Record the results of the inspection and note whether or not the rock socket complies with the requirements of Section 1812.	X	—

1705.7.3 Cast-in-place deep foundation elements. The installation and testing of driven or drilled cast-in-place deep foundation elements shall be subject to special inspections as specified in Tables 1705.7, 1705.7.1, 1705.7.2 and 1705.7.3, as applicable.

**TABLE 1705.7.3
REQUIRED SPECIAL INSPECTIONS OF DRIVEN OR DRILLED CAST-IN-PLACE DEEP FOUNDATION ELEMENTS**

INSTALLATION	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION
1. Prior to placing concrete, inspect the inside of the pile for alignment and damage. Record the results of the inspection, noting whether or not the pile is in compliance with Section 1812 and the construction documents.	X	—
2. Witness the placement of concrete or grout inside the deep foundation element, and record the volume placed. Note the date and time of placement, and whether or not the concrete or grout was placed in compliance with Section 1812 and the construction documents.	X	—

1705.7.4 Additional inspection. For methods of deep foundation element installation not covered in Section 1705.7, perform an additional inspection as required by the approved construction documents and record the results noting whether or not the deep foundation element installation complies with the requirements of Sections 1810, 1811, and 1812, and the construction documents. For steel deep foundation elements, perform an additional inspection in accordance with Section 1705.2. For concrete or grout deep foundation elements, and concrete or grout-filled deep foundation elements, perform an additional inspection in accordance with Section 1705.3. For concrete or grout, and concrete or grout filled deep foundation elements, perform an additional inspection in accordance with Section 1705.3.

1705.7.5 Submittal of records of special inspections. The special inspection agency shall submit to the commissioner records of the inspection of the installation of each deep foundation element. The special inspection agency shall also submit records of testing of deep foundation elements at the request of the commissioner. The submittal shall include the following:

1. A location plan showing the designation identifying each deep foundation element, and the locations of required testing, where applicable.
2. A written summary of the installation and testing performed, signed by the special inspector, including a statement verifying that all installation and testing complies with the requirements of Sections 1810, 1811 and 1812, and the construction documents. The special inspector shall document the type and results of tests, and information of the technician and testing agency performing the test.

1705.8 Prestressed rock and soil anchors installation and testing and grout inspection. Special inspection for the installation and testing of permanent prestressed rock and soil anchors shall be performed in accordance with Section

SPECIAL INSPECTIONS AND TESTS

1705.8.1 and Table 1705.8.1. Special inspection for the installation and testing of grout inspection shall be performed in accordance with Section 1705.8.2.

1705.8.1 Prestressed rock and soil anchors. The installation and testing of permanent prestressed rock and soil anchors shall be subject to special inspections. A special inspector shall be present during the installation of each anchor, including all drilling, grouting, tendon splicing (including corrosion protection) and placement, and installation of any accessories such as trumpets and bearing plates. The special inspector shall also be present for all required load testing and shall create a report containing a log for each element installed and for each load test performed. The special inspection agency shall submit to the commissioner records of the installation of each rock anchor, and where testing is required, the special inspection agency shall also submit records of the testing. The special inspection agency shall submit to the commissioner records of the installation and testing of each anchor.

**TABLE 1705.8.1
REQUIRED SPECIAL INSPECTIONS OF PRESTRESSED SOIL AND ROCK ANCHORS**

INSTALLATION	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION
1. Special inspection of Casing Installation. Special inspector shall observe casing installation and confirm that anchors are installed through a fully cased borehole or installed through stable rock in an open hole drill hole.	X	—
2. Special inspection of Anchor Materials. Special inspector shall confirm that the materials used in prestressed rock and soil anchors conform Section 1815.3 and approved construction documents. Inspector shall confirm that bar tendons conform to ASTM A722 and that strand tendons conform to ASTM A416 or A882.	X	—
3. Special inspection of Anchor Installation. Special inspector shall confirm the preparation and installation of each anchor, including placement, type, size and location of anchors, accessories, corrosion protection and grout conforms to PTI DC35.1-4 and the approved construction documents.	X	—
4. Special inspection of Grout. At the time of grout placement, special inspector shall prepare 3" diameter grout cylinders at the intervals specified in Section 1812.1.1.2.1. Cylinders shall be prepared in accordance with ASTM C31/C31M.	X	—
5. Special inspection of Grout. At the time of grout placement, special inspector shall determine the specific gravity of grout in accordance with American Petroleum Institute (API) Recommended Practice 13B-1 or in accordance with ASTM C138.	X	—
6. Special inspection of Grout. The compressive strength of grout cylinders shall be verified using 3" diameter cylinders tested in accordance with ASTM C39 and Section 1812.1.1.2.2.	X	—
7. Special inspection of Load Test. The special inspector shall observe load testing of each anchor and confirm that the test is accordance with PTI DC35.1-14 (Sections 8.1 – 8.9) and Section 1815.5. Inspector shall confirm that testing lock-off, and test loads comply with approved construction documents and PTI DC35.1–14 (Sections 8.1 – 8.9).	X	—

1705.8.1.1 Records of anchor element installation. The record of installation of each anchor element shall include, but not be limited to, the following:

1. The identifying designation of the anchor element and the date of installation, including the start and end times.
2. The size, material, details of corrosion protection, and allowable capacity as specified in the contract drawings.
3. Unbonded and bonded lengths of tendons, including strata for the bond zone.
4. The date and time of the witnessing of the placement of grout inside the anchor element, the volume placed, and whether or not the grout was placed in compliance with Section 1815 and contract requirements.
5. Equipment and method used for installation and record the time for penetration from the start of installation until completion.

1705.8.1.2 Records of testing of anchor elements. The records of testing of anchor elements shall include, but not be limited to, the following:

1. The results of the inspection and note the lock-off load.
2. The identifying designation for the element being tested, and the date of the testing, including the start and end time.
3. The method of performing the test, including the equipment being used, as well as the test results, noting whether or not the method of testing and the test results comply with the requirements of Section 1815.

1705.8.2 Grout inspection. Grout inspection shall be in accordance with Sections 1705.3 and 1815.8.

1705.9 Helical pile foundations. Special inspections shall be performed continuously during installation of helical pile foundations. The information recorded shall include installation equipment used, pile dimensions, tip elevations, final depth, final installation torque and other pertinent installation data as required by the registered design professional of record. The approved geotechnical report and the construction documents prepared by the registered design professional and standards established by the commissioner pursuant to Section 28-113.2.2 of the *Administrative Code* shall be used to determine compliance. The special inspection agency shall submit to the department records of the inspection of the installation of helical piles on forms provided by the department.

1705.10 Reserved.

1705.11 Reserved.

1705.12 Special inspections for seismic resistance. Special inspections for seismic resistance shall be required as specified in Sections 1705.12.1 through 1705.12.5, unless exempted by the exceptions of Section 1704.2.

1705.12.1 Designated seismic systems. For structures assigned to Seismic Design Category C or D, the special inspector shall examine designated seismic systems requiring seismic qualification in accordance with Section 13.2.2 of ASCE 7 and verify that the label, anchorage and mounting conform to the certificate of compliance.

1705.12.2 Access floors. Periodic special inspection is required for the anchorage of access floors in structures assigned to Seismic Design Category D.

1705.12.3 Plumbing, mechanical, fuel gas, and electrical components. Periodic special inspection of plumbing, mechanical and electrical components shall be required for the following:

1. Anchorage of electrical equipment for emergency and standby power systems in structures assigned to Seismic Design Category C or D.
2. Installation and anchorage of piping systems designed to carry hazardous materials and their associated mechanical units in structures assigned to Seismic Design Category C or D.
3. Installation and anchorage of ductwork designed to carry hazardous materials in structures assigned to Seismic Design Category C or D.
4. Installation and anchorage of vibration isolation systems in structures assigned to Seismic Design Category C or D where the approved construction documents require a nominal clearance of $\frac{1}{4}$ inch (6.4 mm) or less between the equipment support frame and restraint.
5. Installation and anchorage of stainless steel flexible multiple leg hose assemblies in structures assigned to Seismic Design Category C or D.

1705.12.4 Storage racks. Periodic special inspection is required for the anchorage of storage racks that are 8 feet (2438.4 mm) or greater in height in structures assigned to Seismic Design Category D.

1705.12.5 Seismic isolation systems. Periodic special inspection shall be provided for seismic isolation systems in seismically isolated structures assigned to Seismic Design Category B, C or D during the fabrication and installation of isolator units and energy dissipation devices.

1705.13 Testing for seismic resistance. Testing for seismic resistance shall be required as specified in Sections 1705.13.1 through 1705.13.3, unless exempted from special inspections by the exceptions of Section 1704.2.

SPECIAL INSPECTIONS AND TESTS

1705.13.1 Nonstructural components. For structures assigned to Seismic Design Category B, C, or D where the requirements of Section 13.2.1 of ASCE 7 for nonstructural components, supports or attachments are met by seismic qualification as specified in Item 2 therein, the registered design professional shall specify on the approved construction documents the requirements for seismic qualification by analysis, testing or experience data. Certificates of compliance for the seismic qualification shall be submitted to the commissioner.

1705.13.2 Designated seismic systems. For structures assigned to Seismic Design Category C or D and with designated seismic systems that are subject to the requirements of Section 13.2.2 of ASCE 7 for certification, the registered design professional shall specify on the approved construction documents the requirements to be met by analysis, testing or experience data as specified therein. Certificates of compliance documenting that the requirements are met shall be submitted to the commissioner.

1705.13.3 Seismic isolation systems. Seismic isolation systems in seismically isolated structures assigned to Seismic Design Category B, C, or D shall be tested in accordance with Section 17.8 of ASCE 7.

1705.14 Sprayed fire-resistant materials. Special inspections for sprayed fire-resistant materials applied to floor, roof and wall assemblies and structural members shall be in accordance with Sections 1705.14.1 through 1705.14.7. Special inspections shall be based on the fire-resistance design as designated by the approved construction documents. The tests set forth in this section shall be based on samplings from specific floor, roof and wall assemblies and structural members. Special inspections shall be performed after the rough installation of electrical, automatic sprinkler, mechanical and plumbing systems and suspension systems for ceilings, where applicable.

1705.14.1 Physical and visual tests. The special inspections shall include the following tests and observations to demonstrate compliance with the listing and the fire-resistance rating:

1. Condition of substrates.
2. Thickness of application.
3. Density in pounds per cubic foot (kg/m^3).
4. Bond strength adhesion/cohesion.
5. Condition of finished application.

1705.14.2 Structural member surface conditions. The surfaces shall be prepared in accordance with the approved fire-resistance design and the written instructions of approved manufacturers. The prepared surface of structural members to be sprayed shall be inspected before the application of the sprayed fire-resistant material.

1705.14.3 Application. The substrate shall have a minimum ambient temperature before and after application as specified in the written instructions of approved manufacturers. The area for application shall be ventilated during and after application as required by the written instructions of approved manufacturers.

1705.14.4 Thickness. No more than 10 percent of the thickness measurements of the sprayed fire-resistant materials applied to floor, roof and wall assemblies and structural members shall be less than the thickness required by the approved fire-resistance design, but in no case less than the minimum allowable thickness required by Section 1705.14.4.1.

1705.14.4.1 Minimum allowable thickness. For design thicknesses 1 inch (25.4 mm) or greater, the minimum allowable individual thickness shall be the design thickness minus $\frac{1}{4}$ inch (6.4 mm). For design thicknesses less than 1 inch (25.4 mm), the minimum allowable individual thickness shall be the design thickness minus 25 percent. Thickness shall be determined in accordance with ASTM E 605. Samples of the sprayed fire-resistant materials shall be selected in accordance with Sections 1705.14.4.2 through 1705.14.4.9 of this code.

1705.14.4.2 Floor, roof and wall assemblies. The thickness of the sprayed fire-resistant material applied to floor, roof and wall assemblies shall be determined in accordance with ASTM E 605, making not fewer than four measurements for each 1,000 square feet (92.9 m^2) of the sprayed area in each story or portion thereof.

1705.14.4.3 Cellular decks. Thickness measurements shall be selected from a square area, 12 inches by 12 inches (304.8 mm by 304.8 mm) in size. A minimum of four measurements shall be made, located symmetrically within the square area.

1705.14.4.4 Fluted decks. Thickness measurements shall be selected from a square area, 12 inches by 12 inches (304.8 mm by 304.8 mm) in size. Not fewer than four measurements shall be made, located symmetrically within the square area, including one each of the following: valley, crest and sides. The average of the measurements shall be reported.

1705.14.4.5 Structural framing members. The thickness of the sprayed fire-resistant material applied to structural members shall be determined in accordance with ASTM E 605. Thickness testing shall be performed on not less than 25 percent of the structural members on each floor.

1705.14.4.5.1 Beams and girders. At beams and girders thickness measurements shall be made at nine locations around the beam or girder at each end of one 12-inch (304.8 mm) length.

1705.14.4.5.2 Joists and trusses. At joists and trusses, thickness measurements shall be made at seven locations around the joist or truss at each end of one 12-inch (304.8 mm) length.

1705.14.4.5.3 Wide-flanged columns. At wide-flanged columns, thickness measurements shall be made at 12 locations around the column at each end of one 12-inch (304.8 mm) length.

1705.14.4.5.4 Hollow structural section and pipe columns. At hollow structural section and pipe columns, thickness measurements shall be made at not fewer than four locations around the column at each end of one 12-inch (304.8 mm) length.

1705.14.5 Density. The density of the sprayed fire-resistant material shall not be less than the density specified in the approved fire-resistance design. Density of the sprayed fire-resistant material shall be determined in accordance with ASTM E 605. The test samples for determining the density of the sprayed fire-resistant materials shall be selected as follows:

1. From each floor, roof and wall assembly at the rate of not less than one sample for every 2,500 square feet (232.3 m²) or portion thereof of the sprayed area in each story, samples shall be taken at a distance of no less than 50 LF from another sample or 10 LF from any starting edge of an application of a specific type of material.
2. From beams, girders, trusses and columns at the rate of not less than one sample for each type of structural member for each 2,500 square feet (232.3 m²) of floor area or portion thereof in each story.

1705.14.6 Bond strength. The cohesive/adhesive bond strength of the cured sprayed fire-resistant material applied to floor, roof and wall assemblies and structural members shall not be less than 150 pounds per square foot (psf) (7.18 kN/m²). The cohesive/adhesive bond strength shall be determined in accordance with the field test specified in ASTM E 736 by testing in-place samples of the sprayed fire-resistant material selected in accordance with Sections 1705.14.6.1 through 1705.14.6.3 of this code.

Exception: For high-rise buildings, as described in Section 403 of this code, refer to the requirements for minimum bond strength in Table 403.2.4 of this code.

1705.14.6.1 Floor, roof and wall assemblies. The test samples for determining the cohesive/adhesive bond strength of the sprayed fire-resistant materials shall be selected from each floor, roof and wall assembly at the rate of not less than one sample for every 2,500 square feet (232.3 m²) or part thereof of the sprayed area in each story. Samples shall be obtained from locations no less than 50 LF (15 240 mm) apart or 10 LF (3048 mm) from any starting edge of an application of a specific type of material.

1705.14.6.2 Structural members. The test samples for determining the cohesive/adhesive bond strength of the sprayed fire-resistant materials shall be selected from beams, girders, joists, trusses, columns and other structural members at the rate of not less than one sample for each type of structural member for each 2,500 square feet (232.3 m²) of floor area or portion thereof in each story.

1705.14.6.3 Primer, paint and encapsulant bond tests. Bond tests to qualify a primer, paint or encapsulant shall be conducted when the sprayed fire-resistant material is applied to a primed, painted or encapsulated surface for which acceptable bond-strength performance between these coatings and the fire-resistant material has not been determined. A bonding agent approved by the SFRM manufacturer shall be applied to a primed, painted or encapsulated surface where the bond strengths are found to be less than required values.

1705.14.7 Inspection of existing sprayed fire-resistant materials during alterations in office spaces and spaces classified in Occupancy Group B. In office spaces and spaces classified in Occupancy Group B, where an alteration exposes any required sprayed fire-resistant materials on structural members or where, pursuant to an

SPECIAL INSPECTIONS AND TESTS

alteration, persons are required to enter or access areas in which such sprayed fire-resistant materials are capable of being observed, the existing required sprayed fire-resistant materials shall be subject to special inspection. Such inspection shall require the special inspector to perform a visual inspection and report the existing sprayed fire-resistant materials and conditions to the registered design professional.

1705.15 Mastic and intumescent fire-resistant coatings. Special inspections and tests for mastic and intumescent fire-resistant coatings applied to structural elements and decks shall be in accordance with AWCI 12-B. Special inspections and tests shall be based on provisions of AWCI 12-B and the fire-resistance design as designated in the approved construction documents.

1705.16 Combustible exterior wall coverings. Special inspections shall be required for all exterior wall coverings containing combustible materials installed more than 15 feet (4572 mm) above adjacent finished grades and alterations to existing exterior wall coverings containing combustible materials installed more than 15 feet (4572 mm) above adjacent finished grades. Exterior insulation and finish system (EIFS) applications, metal composite materials (MCM), high-pressure decorative exterior-grade compact laminates (HPL) and exterior wall covering containing combustible veneers, combustible framing, combustible water-resistive barriers and foam plastic insulation installed more than 15 feet (4572 mm) above adjacent finished grades and alterations to existing installations more than 15 feet (4572 mm) above adjacent finished grades shall be inspected in accordance with Section 1705.16.1.

Exceptions:

1. Water-resistive barriers that are the only combustible component in the exterior wall covering satisfying the requirements of Exceptions 1 or 2 of Section 1403.5.1.
2. Type VB construction up to 40 feet (12 192 mm) or less in height above grade.

1705.16.1 Inspection program. The special inspection shall include verification of compliance with the approved construction documents for the following: attachment to structure; component properties; thicknesses, orientation, cavity sizes and location of thermal barriers, fireblocking and cavity closures; installation of waterproofing membranes, weeps, drains, mold prevention features as specified; and conformance with the manufacturers' installation instructions, and NFPA 285 tested assembly design documentation. The special inspector shall verify that individual components required to be labeled in accordance with this code do bear such labels.

1705.16.2 Water-resistive barrier coating. A water-resistive barrier coating complying with ASTM E 2570 requires special inspection of the water-resistive barrier coating when installed over a sheathing substrate. Inspection of water-resistive barrier coating must verify conformance with approved construction documents, manufacturer's installation instructions, and NFPA 285 tested assembly design documentation.

1705.16.3 Thermal barrier. The special inspector shall confirm the installation of the thermal barrier is in accordance with the approved construction documents, manufacturer's installation instructions, and NFPA 285 tested assembly design documentation.

Exception: Foam plastic insulation installation on one-story buildings not required to have thermal barrier complying with Section 2603.4.1.4 of this code.

1705.16.4 Fireblocking. The special inspector shall confirm the installation of fireblocking is in accordance with Chapter 7 of this code, the approved construction documents, manufacturer's installation instructions, and NFPA 285 tested assembly design documentation.

Exception: EIFS systems and exterior wall coverings not required to have fireblocking pursuant to Section 718.2.6.1 of this code.

1705.17 Fire-resistant penetrations and joints. Special inspections for through-penetrations, membrane penetration firestops, fire-resistant joint systems, and perimeter fire barrier systems that are tested and listed in accordance with Sections 714.3.1.2, 714.4.2, 715.3 and 715.4 shall comply with Sections 1705.17.1 and 1705.17.2.

1705.17.1 Penetration firestops. Inspections of penetration firestop systems that are tested and listed in accordance with Sections 714.3.1.2 and 714.4.2 of this code shall be conducted by an approved special inspection agency in accordance with ASTM E 2174.

1705.17.2 Joint systems. Inspection of fire-resistant joint systems that are tested and listed in accordance with Sections 715.3 and 715.4 of this code shall be conducted by an approved special inspection agency in accordance with ASTM E 2393.

1705.18 Testing for smoke control. Testing of smoke control systems shall be witnessed in its entirety by a special inspector in accordance with Sections 1705.18.1 and 909.

Exception: Post-fire smoke purge systems that are not required to function as a smoke control system shall be permitted to be inspected pursuant to the special inspection requirements of Section 1705.21.

1705.18.1 Testing scope. The test scope shall be as follows:

1. During erection of ductwork and prior to concealment for the purposes of leakage testing and recording of device location.
2. Prior to occupancy and after sufficient completion for the purposes of pressure difference testing, flow measurements and detection and control verification.

1705.19 Vertical masonry foundation elements. Special inspection shall be performed in accordance with Section 1705.4 for vertical masonry foundation elements.

Exception: Piers for support of lightweight fences, recreational equipment, site furnishings and similar construction, unless special inspection is specifically noted as required on the approved construction documents.

1705.20 Wall panels, curtain walls and veneers. Special inspection is required for exterior architectural wall panels, curtain walls and the anchoring of veneers for the full height of the installation when any portion of the wall panel, curtain wall and veneer installation exceeds an elevation of 40 feet (12 192 mm) above grade. Special inspection of masonry veneer on such structures shall be in accordance with Section 1705.4. Special inspection of combustible exterior wall covering shall be in accordance with Section 1705.16.

Exceptions: Special inspection of wall panels, curtain walls and the anchoring of veneers is not applicable to:

1. Repairs and replacement in kind of gaskets or seals;
2. Reglazing other than 4-sided structural silicone glazing for which a permit is required; or
3. Where exempted by the rules of the department.

1705.20.1 Design and installation documents. The special inspector shall become familiar with and retain a copy of the approved construction documents, and the following items, as applicable, approved by the registered design professional of record:

1. Shop drawings.
2. Instructions for the sequence of component installation.
3. Samples and/or mock-ups, if supplied.
4. Required fire-resistance rating and performance standard listings, as applicable.

1705.20.2 Inspection program. The special inspector shall field check the site conditions at the time the structure is prepared for component installation, and periodically during component installation, to verify the following work is performed in compliance with the approved construction documents, including that:

1. The supporting structure for components being installed is aligned and within specified tolerances required for the components;
2. Required inserts are installed;
3. Framing components are installed and aligned as specified, and without structural defects or weakness;
4. Anchors are placed, welded, bolted and finished as specified, as applicable;
 - a. Inspection of welding of anchors shall be inspected in accordance with Section 1705.3. Inspection of bolting of anchors shall be in accordance with Section 1705.4. Inspection of the installation of post-installed anchors shall be in accordance Section 1705.37.

SPECIAL INSPECTIONS AND TESTS

5. Weeps, flashings and tubes are installed as specified and functioning;
6. Joinery and end dams are sealed as specified;
7. Sealing materials with specified adhesive and movement capabilities are installed;
8. Gaskets, tapes, seals, insulation, flashing and other materials that are barriers to air and water movement, vapor drive, and heat loss are installed as specified;
9. Joint filler materials that accommodate specified horizontal and vertical movement are installed in accordance with the manufacturers' instructions; and
10. Fireblocking and thermal barrier within combustible exterior wall coverings are installed as specified.

1705.21 Special inspection for mechanical systems. Mechanical systems that require a certificate of compliance in accordance with Section 28-116.4.1 of the *Administrative Code* shall be inspected for conformance with the approved construction documents.

1705.21.1 Tests for mechanical systems. Tests of mechanical systems shall be performed in accordance with applicable sections of the code including Sections 507.6, 1011, 1108 and 1208 of the *New York City Mechanical Code*. The special inspector shall witness all required tests.

1705.21.2 Factory built equipment. Listed and labeled self-contained factory-built equipment and appliances deemed by the registered design professional to meet the design, manufacture and factory test requirements of this code shall be field tested in accordance with the manufacturers' installation and operational test standards.

1705.21.3 Special inspection of mechanical systems. Inspections of mechanical systems shall include the following as applicable to the system:

1. Visual certification that required components of such systems are complete in accordance with the manufacturers' installation guidelines and the approved construction documents;
2. Supports, hangers, and vibration isolation equipment are properly spaced and anchored to supporting structure;
 - 2.1. Seismic bracing shall be inspected in accordance with Section 1705.12.
3. Installation of required signage and safety instructions;
4. Electrical components are installed and electrical sign-off issued;
5. Ventilation balancing report is complete and in accordance with design documents and the system is operating as designed;
6. Required labeling, operational instructions and safety signage properly posted;
7. All related special inspections for such systems are complete;
8. The installation of exterior mechanical equipment for compliance with the approved construction documents, including, as applicable:
 - 8.1. Equipment specified;
 - 8.2. Equipment location;
 - 8.3. Installation and attachment details;
 - 8.4. Vibration isolation installation; and
 - 8.5. Other items specified by the applicant of record for the mechanical equipment design necessary to verify compliance with Section 313.2.1 of the *New York City Mechanical Code*.
9. Fire and smoke dampers are installed and functioning properly.

1705.22 Special inspection for fuel storage and fuel-piping systems. Fuel-oil storage equipment, including tanks, pumps, valves, transfer, return, fill and vent piping, hangers and bracing, fill and vent terminals, and related systems regulated by Section 1305 of the *New York City Mechanical Code*, shall be inspected for conformance with the approved construction documents and the manufacturers' installation standards. Testing of fuel storage tanks and piping

shall be performed in accordance with Section 1305 of the *New York City Mechanical Code*. Special inspection of seismic bracing shall be performed in accordance with Section 1705.12 of this code.

Exceptions:

1. Listed fuel oil storage tanks 660 gallons (2498.4 L) or less in capacity, supplied with a label or manufacturer's certification attesting that the tank was factory tested to a testing standard indicated in Section 1308.1 of the *New York City Mechanical Code*.
2. Systems and equipment exempt from service equipment certificate of compliance in accordance with Section 28-116.4.1 of the *Administrative Code*.

1705.23 High-pressure-steam piping and high temperature hot water piping. High-pressure-steam piping and high temperature hot water piping regulated by Section 1211 of the *New York City Mechanical Code*, shall be subject to special inspection in accordance with this section. Special inspection of seismic bracing shall be performed in accordance with Section 1705.12 of this code.

1705.23.1 Welding. The special inspector shall verify the qualifications of the welder and the quality of the welding materials and equipment prior to welding operations. The special inspector shall review the proposed welding procedures for compliance with applicable standards listed in Section 1211.2.2 of the *New York City Mechanical Code*.

1705.23.2 Welding operations. The special inspector shall perform periodic inspection of the pipe joining and welding operations in accordance with the above. Radiographic testing shall be applied to connections as indicated in Section 1211.2.2 of the *New York City Mechanical Code*. Unacceptable connections and installations shall be rejected.

1705.23.3 Testing of high-pressure-steam systems and high temperature hot water piping systems. Hydrostatic testing shall be performed on the completed installation of new and altered systems in accordance with Section 1211.2.3 of the *New York City Mechanical Code*. The special inspector shall witness all required hydrostatic tests.

Exception: Testing requirements shall not apply to alteration or repairs to existing piping systems in which the integrity of the piping is not affected. Such alterations shall be visually inspected for compliance with the approved construction documents.

1705.24 High-pressure-gas piping. High-pressure-gas piping regulated in Section 406 of the *New York City Fuel Gas Code* shall be subject to special inspection in accordance with this section. Special inspection of seismic bracing shall be performed in accordance with Section 1705.12 of this code.

1705.24.1 Welding. The special inspector shall verify the qualifications of the welder, the quality of the welding materials and equipment prior to welding operations. The special inspector shall review the proposed welding procedures for compliance with applicable standards listed in Section 406 of the *New York City Fuel Gas Code*.

1705.24.2 Welding operations. The special inspector shall perform periodic special inspection for the pipe joining and welding operations in accordance with the above. Radiographic testing shall be performed to the connections as indicated in Section 406 of the *New York City Fuel Gas Code*.

1705.24.3 Testing of high-pressure-gas piping. Pressure testing shall be performed on the completed installation of new and altered systems in accordance with Section 406 of the *New York City Fuel Gas Code*. The special inspector shall witness all required pressure tests.

1705.25 Structural stability. Special inspection for structural stability is an inspection of a structure to verify the ability of such structural system to remain in position, or revert to the original position or another stable equilibrium position acceptable by this code, without incurring damage to the structural system from the activity or load to which it had been subjected until the completion of construction. Such special inspection shall be required for construction work as specified in this section or elsewhere in this code. Structural materials and methods of construction utilized in temporary protections shall be subject to special inspection when such materials and methods of construction would be subject to special inspection as a permanent installation in accordance with the applicable sections of this chapter, including but not limited to special inspections for concrete, structural steel, and deep foundation installation. The registered design professional responsible for plans for a new building, alteration, or other work requiring structural

SPECIAL INSPECTIONS AND TESTS

stability inspections shall identify those areas on the plans submitted to the department in accordance with Section 1704.1.1.1. The means and methods of implementing the structural stability measures shall be prepared by a registered design professional and filed with the department where required in this section and elsewhere in this code.

Exception: Alterations consisting of the replacement of existing exterior lintels located less than 75 feet (22 860 mm) above curb level spanning less than 4 feet (1220 mm) and existing interior headers spanning less than 4 feet (1220 mm), provided the size of the existing span is not increased, shall be exempt from special inspection for structural stability.

1705.25.1 Alterations to existing structures. Alterations to existing structures in which loads are transferred from one structural system of structural elements to another, such as installation of columns or girders, replacement of existing bearing walls, the creation of openings or slots in existing walls, girders or floors, alteration of arches, rigid frames, trusses in frame buildings, where the stability or integrity of a structural system is to be temporarily diminished, or where otherwise required by the commissioner, shall be subject to special inspections in accordance with Sections 1705.25.7 through 1705.25.10.

1705.25.2 Construction operations influencing adjacent structures. Where construction operations have the potential to affect structurally the condition or occupancy of the subject structure and/or an adjacent structure, the structural stability of such structures shall be subject to special inspections in accordance with Sections 1705.25.7 through 1705.25.10.

‡‡‡ **1705.25.3 Excavations.** Methods employed to protect the sides of excavations that require construction documents in accordance with Section 3304.4.1, and blasting for the purpose of excavation shall be subject to special inspections in accordance with Sections 1705.25.7 through 1705.25.10.

1705.25.3.1 Slurry. The proportions and installation of slurry mixtures to protect the sides of excavations shall be subject to special inspection. Slurry mix proportions and installation procedures shall be provided by a registered design professional, and the special inspection agency shall verify compliance with the approved procedures.

1705.25.3.2 New foundations. In addition to the special inspection for structural stability, any new foundation elements installed as part of support of excavation operations shall be subject to special inspection as a permanent installation in accordance with the applicable sections of this chapter, including, but not limited to, special inspection for concrete, structural steel, and deep foundation installation.

1705.25.3.3 Effect of blasting. Adjoining properties shall be subject to special inspection for structural stability due to the effects of blasting when blasting is utilized for excavation.

1705.25.4 Underpinning and alternate methods of support of buildings and adjacent property. Underpinning of structures and alternate methods of support of buildings and adjacent property shall be subject to special inspections in accordance with Sections 1705.25.7 through 1705.25.10.

1705.25.4.1 New foundations. In addition to the special inspection for structural stability, any new foundation elements installed as part of underpinning operations shall be subject to special inspection as a permanent installation in accordance with the applicable sections of this chapter, including, but not limited to, special inspection for concrete, structural steel, and deep foundation installation.

1705.25.5 Demolition. Demolition operations shall be subject to special inspection in accordance with Sections 1705.25.5.1 and 1705.25.5.2 as applicable to the operations.

1705.25.5.1 Mechanical means and methods of demolition. Where mechanical demolition equipment, other than handheld devices, is to be used in the full or partial demolition of a building from within the building, or is to be used within the building to remove debris or move material, such demolition operation shall be subject to special inspection in accordance with Sections 1705.25.7 through 1705.25.10. The special inspector shall visit the site a minimum of three times: before demolition operations start, during demolition, and at the conclusion of demolition. Additionally, the special inspector shall visit the site at minimum intervals identified in the design documents and set forth in Section 1705.25.8.2.

1705.25.5.2 Other means of demolition. Where full or partial demolition of a weakened structure as specified in Section 3306.7 or the demolition of a masonry wall with a height to thickness ratio of 16 or greater is proposed, or the demolition of a floor structure directly supported by such wall, such demolition operation shall be subject to special inspection in accordance with Sections 1705.25.7 through 1705.25.10. The special inspector

shall visit the site a minimum of three times: before demolition operations start, during demolition, and at the conclusion of demolition. Additionally, the special inspector shall visit the site at minimum intervals identified in the design documents and set forth in Section 1705.25.8.2.

1705.25.6 Raising and moving of a building. A periodic special inspection shall be required in accordance with Sections 1705.25.7 through 1705.25.10 where the lowest above-grade floor or the lowest subgrade floor of a building is to be raised, lifted, elevated or moved.

1705.25.7 Inspection program. Prior to commencement of work requiring special inspection per Section 1705.25, the special inspector shall review the approved construction documents, related reports, and contractor's sequence of operations and confirm that all areas of work requiring special inspection per Section 1705.25 have design documents addressing the structural stability temporary protections and sequence. A written statement shall be included in the design documents prepared in accordance with Section 1705.25.8, indicating:

1. The portions of work requiring design documents in accordance with Section 1705.25.8.
2. The names and addresses of the licensed professionals that have been engaged to supply design documents for applicable work.
3. Where filing of plans is required, the application and permit numbers for the design documents for the applicable work.
4. A schedule of special inspections, at agreed intervals, including adequate frequency to assure the contractor's continued compliance with the proposed designs and sequence of construction operations. At a minimum, the site must be inspected twice, once at a pre-construction meeting with the contractor and once during construction operations.

1705.25.8 Design documents. Design documents, including shop drawings, sketches and written descriptions of proposed work regarding structural stability in construction operations shall be prepared by a registered design professional in the employ of the owner or the contractor. Where required by this section or elsewhere in this code, the work associated with the structural stability protection methods shall be filed with the department. When filing of the design of protection methods is not required, such designs may be revised at any time by the registered design professional. Copies of the special inspection log book, the design documents and revisions thereof, shall be maintained at the job site, and at the office of the special inspector, available for use and review at all reasonable times, until the work subject to special inspection is complete. In the case of alteration to existing structures, the structural stability design documents shall be reviewed by the registered design professional of record.

1705.25.8.1 Monitoring. The design documents for structural stability shall include any requirements for monitoring of the subject structure and/or adjacent structures, as determined by the registered design professional responsible for the design. The monitoring plan shall be specific to the buildings to be monitored and operations to be undertaken, and shall specify the scope and frequency of monitoring, acceptable tolerances, and reporting criteria for when tolerances are exceeded, including when the department is required to be notified. The monitoring plan shall include provisions for daily monitoring, at a minimum, until the structural stability protection methods are no longer necessary. Such monitoring can be accomplished by remote means, in-person monitoring or any combination thereof.

1705.25.8.1.1 Additional monitoring requirements for demolition. The design documents for structural stability shall identify and specify the monitoring required for the demolition operation and be specific to all phases of the demolition operation. These specifications must be part of the requirements identified for special inspection.

1705.25.8.2 Additional requirements for design documents for demolition. Design documents for demolition shall indicate a project-specific required minimum interval for special inspections. A written statement of what the special inspection must include shall be provided on the design documents. The minimum special inspection interval must be incorporated into the demolition sequence. Building components and phases of demolition deemed to be structurally critical must be identified for inspection. A monitoring protocol must be provided in accordance with the requirements of Section 1705.25.8.1.1.

1705.25.8.3 Additional requirements for design documents for underpinning and alternate methods of support of buildings and adjacent property. Design documents for underpinning and alternate methods of support of buildings and adjacent property shall comply with the requirements of Section 1817. Such drawings

SPECIAL INSPECTIONS AND TESTS

shall indicate a project-specific required minimum interval for special inspections. A written statement of what the special inspection must include shall be provided on the design documents. The minimum special inspection interval must be incorporated into the sequence of operations. Building components and phases of operations deemed to be structurally critical must be identified for inspection. A monitoring protocol must be provided in accordance with the requirements of Section 1705.25.8.1.1. At a minimum, underpinning shall be subject to continuous special inspection for structural stability of the structure being underpinned while work is occurring.

1705.25.8.4 Additional requirements for design documents for protection of the sides of excavations and blasting. Design documents for protection of the sides of excavations and blasting methods of support of buildings and adjacent property shall indicate a project-specific required minimum interval for special inspections. A written statement of what the special inspection must include shall be provided on the design documents. The minimum special inspection interval must be incorporated into the sequence of operations. Building components and phases of operations deemed to be structurally critical must be identified for inspection. A monitoring protocol must be provided in accordance with the requirements of Section 1705.25.8.1.1. At a minimum, protection of the sides of excavations and blasting work shall be subject to continuous special inspection for structural stability of the structure or adjacent property being affected while work is occurring.

††† **1705.25.9 Inspection during construction operations.** The special inspector shall visit the jobsite at agreed intervals, assess the ongoing work and verify that operations conform with the design documents. Deficiencies shall be reported as required by Section 1704.1.1.2. In the event unsafe conditions are discovered, the commissioner and the registered design professional employed by the contractor shall be immediately notified by the special inspector.

1705.25.10 Records of special inspections. The special inspection logbook shall be maintained at the special inspector's office and shall contain the following information:

1. Project identification, application number and address.
2. Date and time of each inspection.
3. Names of personnel who performed each inspection.
4. Dates of on-site and off-site meetings, names of the participants and a summary of the conversations.
5. Any significant observations or instructions given related to structural stability in any of the following:
 - 5.1. Deviations from the design documents.
 - 5.2. Anticipated field conditions.
 - 5.3. Proper execution of the work.
 - 5.4. Safe jobsite conditions.
 - 5.5. Precautions taken to maintain safe conditions, if work is stopped for any reason.

1705.26 Tenant protection plan compliance. When alteration, partial demolition, or construction operations are performed at occupied multiple dwellings, the special inspector shall periodically verify compliance with a tenant protection plan as provided for in Chapter 1 of Title 28 of the *Administrative Code* and Section 3303.10 of this code. Special inspections for compliance with the tenant protection plan shall comply with Sections 1705.26.1 and 1705.26.2 of this code.

Exceptions: Special inspection for tenant protection plan compliance is not required where the scope of the work is limited to:

1. A three-family home;
2. The interior of a single dwelling unit of an occupied multiple dwelling with no disruption to the essential services of other units, where such dwelling is owner-occupied. For a dwelling unit within a property that is owned by a condominium or held by a shareholder of a cooperative corporation under a proprietary lease, the unit must be occupied by the owners of record for such unit; or
3. The interior of a single dwelling unit of an occupied multiple dwelling three stories or less in height.

1705.26.1 Inspection frequency. Special inspections shall be performed throughout the duration of the tenant protection plan at sufficient periodic intervals to verify compliance with the tenant protection plan throughout the course of work. At a minimum, inspections shall be performed:

1. Prior to the start of construction and/or demolition to document conditions prior to work;
2. At the start of construction and/or demolition;
3. Once per week during construction and/or demolition;
4. After a tenant protection plan violation has been issued to verify that the violation was corrected;
5. When the location of the alteration or construction operations performed in conjunction with the tenant protection plan has moved to another location; and
6. Whenever construction or demolition operations have changed, requiring changes in methods of protection.

1705.26.2 Reporting. A written record of the special inspections performed for the tenant protection plan shall be maintained in a log prepared by the special inspector. The log shall include an entry for each inspection and shall indicate that the alteration or construction operation is being performed in accordance with the tenant protection plan. Copies of the special inspection log book shall be maintained at the job site, and at the office of the special inspector, and shall be available for use and review at all reasonable times, until the work subject to special inspection is complete.

1705.27 Private on-site storm water disposal systems and detention facilities. Storm water detention and retention systems required to comply with Section 1114 of the *New York City Plumbing Code*, and roof detention facilities shall be inspected for compliance with the provisions of this section.

1705.27.1 Subsurface investigation special inspection. Soil borings, testpits and soil percolation testing shall be subject to the special inspection requirements of this section.

1705.27.1.1 Soil borings and testpits. Prior to the approval of construction documents, soil borings and testpits shall be performed in accordance with the provisions of Section 1114.2.1 of the *New York City Plumbing Code*. Soil borings and testpits shall be performed under the supervision of a special inspector. The results of the soil borings and testpits shall be filed on forms provided by the department.

1705.27.1.2 Soil percolation tests. Prior to the approval of construction documents, soil percolation tests shall be performed in accordance with the provisions of Section 1114.2.2 of the *New York City Plumbing Code* at the site of a proposed on-site stormwater drainage and detention facility installation to determine the suitability of the soil and site. Such test shall be performed under the supervision of a special inspector. The results of the percolation tests shall be filed on forms provided by the department, stating the suitability of the site and the capacity of the subsoil for the proposed use. The registered design professional of record for the dry-well system shall be notified immediately if the results of the percolation test demonstrate that the system may not function as designed.

1705.27.2 Installation special inspection. Private on-site stormwater disposal systems and detention facilities shall be inspected for conformance with the approved construction documents. Minor variations, based on actual site conditions, shall be acceptable at the discretion of the registered design professional of record. Verification shall include:

1. Materials of construction.
2. Compliance of bedding, per Section 306 of the *New York City Plumbing Code*, of pipe and facilities.
3. Placement and installation of fill materials.
4. Volume of drywell and detention facilities.
5. Installation of volume flow control devices.
6. Workmanship and quality of installation.
7. Overall conformance with the approved construction documents.

SPECIAL INSPECTIONS AND TESTS

8. Detention tanks, and roofs used for stormwater detention, shall be filled with water to demonstrate water tightness for a 2-hour duration or as specified by the registered design professional or the manufacturer's specifications.

1705.28 Individual on-site private sewage disposal systems. Individual on-site private sewage disposal systems shall be inspected in accordance with the rules of the department.

1705.29 Sprinkler system special inspection. New and altered sprinkler systems shall be inspected in accordance with Sections 903 and 1705.29.1. The permit holder responsible for the sprinkler work shall perform all required acceptance tests, complete and sign the appropriate contractor's material and test certifications. The special inspector shall witness all required tests and shall verify that all installations of all materials, fittings, hangers, assemblies and signage are in accordance with the approved construction documents, that painting of the sprinkler system required by Section 903.6 of this code has been performed and that the contractor has transmitted required maintenance literature and instruction to the owner. The special inspector shall verify that the material and test certification forms have been transmitted to the Fire Department and the Department of Buildings. Seismic bracing shall be inspected in accordance with Section 1705.12.

Exception: If the department witnesses a hydrostatic pressure test, it need not be witnessed by the special inspection agency.

1705.29.1 Hydrostatic pressure testing. All new or altered sprinkler systems in buildings shall undergo successful hydrostatic pressure testing by a licensed master plumber or licensed fire suppression piping contractor in accordance with the requirements of this code, including Section 901.5 of this code, and NFPA 13.

Exceptions: Modification of six or fewer sprinklers:

1. May be tested under working pressure in lieu of hydrostatic testing, or
2. Need not be witnessed by a special inspection when the hydrostatic test is performed by the licensed installing contractor.

††† **1705.30 Standpipe system special inspection.** New and altered standpipe systems shall be inspected and tested in accordance with Sections 905 and 1705.30. Fire pump tests associated with standpipe systems shall be tested in accordance with Section 1705.30.2. The permit holder responsible for the standpipe work shall perform all required acceptance tests, and complete and sign the appropriate contractor's material and test certifications. The special inspector shall witness all required tests, verify that installation of all materials, fittings, hangers, assemblies and signage are in accordance with the approved construction documents, that painting of the standpipe system required by Section 905.11 has been performed and that the contractor has transmitted required maintenance literature and instruction to the owner. The special inspector shall verify that the material and test certification forms have been transmitted to the Fire Department and the Department of Buildings. Seismic bracing shall be inspected in accordance with Section 1705.12.

Exception: The special inspection agency need not witness the hydrostatic pressure test when such test is witnessed by the department.

1705.30.1 Hydrostatic pressure testing. All new or altered standpipe systems in buildings shall undergo successful hydrostatic pressure testing by a licensed master plumber or licensed fire suppression piping contractor in accordance with the requirements of this code, including Section 901.5, Sections 1705.30.1.1 through 1705.30.1.5 and NFPA 14.

Exception: When the standpipe system is exposed to freezing conditions, a hydrostatic pressure test required by this section may be postponed until such conditions no longer exist, notwithstanding any requirement that the standpipe be maintained in a state of readiness, provided that the system undergoes an interim test with dry nitrogen or air using a compressor in accordance with NFPA 14. Any such air pressure tests shall be witnessed by the special inspector unless witnessed by the department.

1705.30.1.1 New buildings under construction. For standpipes required to comply with Section 3303.8 of this code, an initial hydrostatic pressure test of the entire system shall be performed when the building reaches a height of 75 feet (22 860 mm) and additional successful hydrostatic pressure tests of the entire system shall be performed at 175 feet (53 340 mm), and at every 100 feet (30 480 mm) in height thereafter. The permit holder

shall perform a final acceptance test of the completed system in accordance with the requirements of Section 901.5.

1705.30.1.2 Enlargements or additions to existing system. Where there is an enlargement that triggers a new standpipe system or there is an addition to an existing standpipe system, hydrostatic pressure tests of the entire system shall be performed for every 75 feet (22 860 mm) of additional height added to the system. The permit holder shall perform a final acceptance test of the completed system in accordance with the requirements of Section 901.5.

1705.30.1.3 Removal of stories, including full demolitions. Where stories are removed from a building served by an existing standpipe system, hydrostatic pressure tests of the entire system shall be performed prior to the commencement of work.

1705.30.1.4 Alterations. For alterations not covered under Sections 1705.30.1.2 or 1705.30.1.3 above, the permit holder shall perform a final acceptance test of the completed system in accordance with the requirements of Section 901.5.

1705.30.1.5 Readiness. No standpipe system shall be considered in readiness until there has been a successful hydrostatic pressure test.

1705.30.2 Fire pump tests. Fire pump tests associated with standpipe systems shall be tested in accordance with Section 913.5 of this code. The special inspector shall witness all required tests and shall verify that all installations comply with NFPA 20 and are in accordance with the approved construction documents.

1705.31 Heating systems. Special inspection shall be required for new and altered boilers and heating systems. All boilers and heating systems, including chimney connectors, shall be inspected for compliance with the approved construction documents. New heating systems shall be tested in accordance with Section 1011 of the *New York City Mechanical Code*. Alterations to heating systems shall be subjected to applicable tests for the altered portions of the system and to verify its satisfactory operation within the existing system. Special inspection of seismic bracing shall be performed in accordance with Section 1705.12 of this code.

Exception: Tests and inspections need not duplicate any valid tests or inspections previously certified by the commissioner or a duly authorized insurance company.

1705.32 Chimneys and vents. New and altered chimneys and vents, replacing an existing appliance in accordance with Section 503.5.6 of the *New York City Fuel Gas Code*, or change in appliance in accordance with Section 507 of the *New York City Fuel Gas Code* and Section 807 of the *New York City Mechanical Code* shall be subject to special inspection. The chimney or vent shall be inspected to verify compliance with the approved construction documents, and proper clearance or isolation from adjacent combustible construction. Testing of the chimney or vent shall be performed as applicable in accordance with Section 810 of the *New York City Mechanical Code* and Section 503.5.6 of the *New York City Fuel Gas Code*. Special inspection of seismic bracing shall be performed in accordance with Section 1705.12 of this code.

Exception: A pressurized smoke test need not be performed on existing and new Listed Type B gas vents and Type L vents. Listed Type B gas vents and Type L vents shall be visually inspected for the presence of a continuous inner wall, proper installation, and no damage or deterioration via readily accessible areas of the vent system, such as the appliance, tee caps, cleanout openings, or termination. The vent system shall be inspected for proper clearances, protection and damage.

1705.32.1 Reuse of existing chimneys. When a new heating system appliance is installed, the special inspector shall verify the condition of the existing chimney lining and breaching. Deficiencies shall be reported as required by Section 1704.1.1.2.

1705.33 Aluminum construction. The special inspections for structural aluminum elements of buildings and structures shall be as required by Section 1705.33.1.

1705.33.1 Welding operations. All welding operations of aluminum elements shall be subject to special inspection for compliance with this code, AA ADM and AWS D1.2.

Exception: Welding operations in connections where the calculated stresses in the welds are less than 50 percent of the basic allowable values. Such connections shall be specifically indicated on the approved construction documents.

SPECIAL INSPECTIONS AND TESTS

1705.34 Flood zone compliance. Special inspection for flood zone compliance shall be as required by Appendix G of this code.

1705.35 Luminous exit path markings. The installation of luminous exit path markings shall be subject to special inspection in accordance with the requirements of Section 1025.8. The special inspector shall verify that approved luminous exit path markings were provided and installed in accordance with Section 1704, the approved construction documents, and manufacturer's instructions.

1705.36 Emergency and standby power systems (generators). The installation of generators as part of emergency and standby power systems shall be subject to special inspection. The special inspector shall: perform visual inspections; verify that the installations of all materials, equipment and signage complies with the installation standards; and verify that the installation of the generator and associated connections complies with the approved construction documents. The special inspector shall witness tests that verify the automatic operation of the emergency or standby power system in accordance with NFPA 110 and NFPA 111. Deficient results shall be rejected, and deficiencies shall be corrected and successfully retested in the presence of the special inspector prior to sign-off of the emergency or standby power system special inspection. Special inspection of seismic bracing shall be performed in accordance with Section 1705.12 of this code.

1705.37 Post-installed anchors. The installation of post-installed mechanical anchors, adhesive anchors, and screw anchors shall comply with Table 1705.37. The special inspection shall include the verification of compliance with approved construction documents and standards established by the commissioner pursuant to Section 28-113.2.2 of the *Administrative Code*.

**TABLE 1705.37
REQUIRED VERIFICATION AND INSPECTION OF POST-INSTALLED ANCHORS**

TYPE	CONTINUOUS SPECIAL INSPECTION	PERIODIC SPECIAL INSPECTION	REFERENCED STANDARD	BC REFERENCE
1. Preparation, placement, type, size and location of anchors, including other details of anchors, installed in hardened concrete, and installed to hardened concrete and to another construction.				1901.1
a. Adhesive anchors installed horizontally or upwardly inclined orientations to resist sustained tension loads.	X	—	ACI 318 17.8.2.4, ACI 318-11 D.9.2.4	1901.1
b. Mechanical anchors and adhesive anchors not defined in 1.a.	—	X	ACI 318 17.9.2	1901.1
2. Preparation, placement, type, size and location of anchors, including other details of anchors, installed in masonry, and installed to masonry and to another construction.	—	X	Manufacturer's specifications and installation instructions	Applicable department procedures
3. Preparation, placement, type, size and location of anchors, including other details of anchors, installed in stone, and installed to stone and to another construction.	—	X	Manufacturer's specifications and installation instructions	

1705.38 Special inspection for nonpotable water systems. Testing and inspection of nonpotable water systems shall be in accordance with Chapter 13 of the *New York City Plumbing Code*. The special inspector shall witness all required tests in accordance with Chapter 13 of the *New York City Plumbing Code*.

1705.38.1 Tests for nonpotable water systems. Tests of nonpotable water systems shall be performed in accordance with Sections 1705.38.1.1 through 1705.38.1.2 of this code.

†††**1705.38.1.1 Water storage tanks.** Water storage tanks shall be tested in accordance with Sections 1302.12.2 and 1303.15.4 of the *New York City Plumbing Code*.

1705.38.1.2 Roofwashers. Roofwashers shall be tested in accordance with Section 1303.15.2 of the *New York City Plumbing Code*.

1705.38.2 Special inspection of nonpotable water systems. Inspections of nonpotable water systems shall include the following as applicable to the system:

1705.38.2.1 Visual inspection that components of such systems are complete in accordance with the manufacturers' instructions and the approved construction documents;

1705.38.2.2 Required labeling, operational instructions and safety signage properly posted.

1705.38.2.3 Inspection of vermin and insect protection. Inlets and vents to the system shall be inspected in accordance with Sections 1302.12.5 and 1303.15.7 of the *New York City Plumbing Code*.

1705.39 Special inspection for exhaust and vent systems in plenums and ducts. Chimneys, vents, and exhaust systems under positive pressure that penetrate ducts or plenums installed in accordance with exceptions 1 and 2 of Section 601.4 of the *New York City Mechanical Code* shall be inspected for conformance with the approved construction documents.

1705.39.1 Visual inspection. Visual inspection shall confirm that joints, seams, and connections of covered mechanical systems are securely fastened and sealed with welds, gaskets, mastics (adhesives), mastic-plus-embedded-fabric systems, liquid sealants or tapes.

1705.39.2 Smoke test. The special inspection shall include witnessing the system smoke test performed in accordance with Section 810 of the *New York City Mechanical Code*.

SECTION BC 1706 DESIGN STRENGTHS OF MATERIALS

1706.1 Conformance to standards. The design strengths and permissible stresses of any structural material that are identified by a manufacturer's designation as to manufacture and grade by mill tests, or the strength and stress grade is otherwise confirmed to the satisfaction of the commissioner shall conform to the specifications and methods of design of accepted engineering practice or the rules of the department in the absence of applicable standards.

1706.2 New materials. For materials that are not specifically provided for in this code, the design strengths and permissible stresses shall be established by tests as provided for in Section 1707.

SECTION BC 1707 ALTERNATIVE TEST PROCEDURE

1707.1 General. In the absence of rules or other approved standards, and upon special application by a registered design professional the commissioner shall make, or cause to be made, the necessary tests and investigations; or the commissioner shall accept duly authenticated reports from approved agencies in respect to the quality and manner of use of new materials or assemblies as provided for in Article 113 of Chapter 1 of Title 28 of the *Administrative Code*. The cost of all tests and other investigations required under the provisions of this code shall be borne by the permit applicant.

SECTION BC 1708 IN-SITU LOAD TESTS

1708.1 General. Whenever there is a reasonable doubt as to the stability or load-bearing capacity of a completed building, structure or portion thereof for the expected loads, an engineering assessment shall be required. The engineering assessment shall involve either a structural analysis or an in-situ load test, or both. The structural analysis shall be based on actual material properties and other as-built conditions that affect stability or load-bearing capacity, and shall be conducted in accordance with the applicable design standard. If the structural assessment determines that the load-bearing capacity is less than that required by the code, load tests shall be conducted in accordance with Section 1708.2. If the building, structure or portion thereof is found to have inadequate stability or load-bearing

SPECIAL INSPECTIONS AND TESTS

capacity for the expected loads, modifications to ensure structural adequacy or the removal of the inadequate construction shall be required.

1708.2 Test standards. Structural components and assemblies shall be tested in accordance with the appropriate material standards listed in Chapter 35. In the absence of a standard that contains an applicable load test procedure, the test procedure shall be developed by a registered design professional and approved. The test procedure shall simulate loads and conditions of application that the completed structure or portion thereof will be subjected to in normal use.

1708.3 In-situ load tests. In-situ load tests shall be conducted in accordance with Section 1708.3.1 or Section 1708.3.2 and shall be supervised by a registered design professional. The test shall simulate the applicable loading conditions specified in Chapter 16 as necessary to address the concerns regarding structural stability of the building, structure or portion thereof.

1708.3.1 Load test procedure specified. Where a standard listed in Chapter 35 contains an applicable load test procedure and acceptance criteria, the test procedure and acceptance criteria in the standard shall apply. In the absence of specific load factors or acceptance criteria, the load factors and acceptance criteria in Section 1708.3.2 shall apply.

1708.3.2 Load test procedure not specified. In the absence of applicable load test procedures contained within a standard referenced by this code or acceptance criteria for a specific material or method of construction, such existing structure shall be subjected to a test procedure developed by a registered design professional that simulates applicable loading and deformation conditions. For components that are not a part of the seismic-load-resisting system, the test load shall be equal to two times the unfactored design loads. The test load shall be left in place for a period of 24 hours. The structure shall be considered to have successfully met the test requirements where the following criteria are satisfied:

1. Under the design load, the deflection shall not exceed the limitations specified in Section 1604.3.
2. Within 24 hours after removal of the test load, the structure shall have recovered not less than 75 percent of the maximum deflection.
3. During and immediately after the test, the structure shall not show evidence of failure.

SECTION BC 1709 PRECONSTRUCTION LOAD TESTS

1709.1 General. In evaluating the physical properties of materials and methods of construction that are not capable of being designed by approved engineering analysis or do not comply with applicable material design standards listed in Chapter 35, the structural adequacy shall be predetermined based on the load test criteria established in this section.

1709.2 Load test procedures specified. Where specific load test procedures, load factors and acceptance criteria are included in the applicable design standards listed in Chapter 35, such test procedures, load factors and acceptance criteria shall apply. In the absence of specific test procedures, load factors or acceptance criteria, the corresponding provisions in Section 1709.3 shall apply.

1709.3 Load test procedures not specified. Where load test procedures are not specified in the applicable design standards listed in Chapter 35, the load-bearing and deformation capacity of structural components and assemblies shall be determined on the basis of a test procedure developed by a registered design professional that simulates applicable loading and deformation conditions. For components and assemblies that are not a part of the seismic-load-resisting system, the test shall be as specified in Section 1709.3.1. Load tests shall simulate the applicable loading conditions specified in Chapter 16.

1709.3.1 Test procedure. The test assembly shall be subjected to an increasing superimposed load equal to not less than two times the superimposed design load. The test load shall be left in place for a period of 24 hours. The tested assembly shall be considered to have successfully met the test requirements if the assembly recovers not less than 75 percent of the maximum deflection within 24 hours after the removal of the test load. The test assembly shall then be reloaded and subjected to an increasing superimposed load until either structural failure occurs or the superimposed load is equal to two and one-half times the load at which the deflection limitations specified in

Section 1709.3.2 were reached, or the load is equal to two and one-half times the superimposed design load. In the case of structural components and assemblies for which deflection limitations are not specified in Section 1709.3.2, the test specimen shall be subjected to an increasing superimposed load until structural failure occurs or the load is equal to two and one-half times the desired superimposed design load. The allowable superimposed design load shall be taken as the lesser of:

1. The load at the deflection limitation given in Section 1709.3.2.
2. The failure load divided by 2.5.
3. The maximum load applied divided by 2.5.

1709.3.2 Deflection. The deflection of structural members under the design load shall not exceed the limitations in Section 1604.3.

1709.4 Wall and partition assemblies. Load-bearing wall and partition assemblies shall sustain the test load both with and without window framing. The test load shall include all design load components. Wall and partition assemblies shall be tested both with and without door and window framing.

1709.5 Exterior window and door assemblies. The design pressure rating of exterior windows and doors in buildings shall be determined in accordance with Section 1709.5.1 or Section 1709.5.2.

Exception: Structural wind load design pressures for window units smaller than the size tested in accordance with Section 1709.5.1 or Section 1709.5.2 shall be permitted to be higher than the design value of the tested unit provided such higher pressures are determined by accepted engineering analysis. All components of the small unit shall be the same as the tested unit. Where such calculated design pressures are used, they shall be validated by an additional test of the window unit having the highest allowable design pressure.

1709.5.1 Exterior windows and doors. Exterior windows and doors shall be tested and labeled as conforming to AAMA/WDMA/CSA 101/I.S.2/A440. The label shall state the name of the manufacturer, the approved labeling agency and the product designation as specified in AAMA/WDMA/CSA 101/I.S.2/A440. Products tested and labeled as conforming to AAMA/WDMA/CSA 101/I.S.2/A440 shall not be subject to the requirements of Sections 2403.2 and 2403.3.

1709.5.2 Exterior windows and door assemblies not provided for in Section 1709.5.1. Exterior window and door assemblies shall be tested in accordance with ASTM E 330. Structural performance of garage doors shall be determined in accordance with either ASTM E 330 or ANSI/DASMA 108, and shall meet the acceptance criteria of ANSI/DASMA 108. Exterior window and door assemblies containing glass shall comply with Section 2403 of this code. The design pressure for testing shall be calculated in accordance with Chapter 16 of this code. Each assembly shall be tested for 10 seconds at a load equal to 1.5 times the design pressure.

1709.6 Skylights and sloped glazing. Skylights and sloped glazing shall comply with the requirements of Chapter 24.

1709.7 Test specimens. Test specimens and construction shall be representative of the materials, workmanship and details normally used in practice. The properties of the materials used to construct the test assembly shall be determined on the basis of tests on samples taken from the load assembly or on representative samples of the materials used to construct the load test assembly. Required tests shall be conducted or witnessed by an approved agency.

SECTION BC 1710 TEST SAFE LOAD

1710.1 Where required. Where proposed construction is not capable of being designed by approved engineering analysis, or where proposed construction design method does not comply with the applicable material design standard, the system of construction or the structural unit and the connections shall be subjected to the tests prescribed in Section 1711. The commissioner shall accept certified reports of such tests conducted by an approved testing agency, provided that such tests meet the requirements of this code and approved procedures.

**SECTION BC 1711
MATERIAL AND TEST STANDARDS**

1711.1 Test standards for joist hangers and connectors. Test standards for joist handlers and connectors shall be in accordance with Sections 1711.1.1 through 1711.2.2.

1711.1.1 Test standards for joist hangers. The vertical load-bearing capacity, torsional moment capacity and deflection characteristics of joist hangers shall be determined in accordance with ASTM D 1761, using lumber having a specific gravity of 0.49 or greater, but not greater than 0.55, as determined in accordance with AF&PA NDS for the joist and headers.

Exception: The joist length shall not be required to exceed 24 inches (609.6 mm).

1711.1.2 Vertical load capacity for joist hangers. The vertical load capacity for the joist hanger shall be determined by testing a minimum of three joist hanger assemblies as specified in ASTM D 1761. If the ultimate vertical load for any one of the tests varies more than 20 percent from the average ultimate vertical load, at least three additional tests shall be conducted. The allowable vertical load of the joist hanger shall be the lowest value determined from the following:

1. The lowest ultimate vertical load for a single hanger from any test divided by three (where three tests are conducted and each ultimate vertical load does not vary more than 20 percent from the average ultimate vertical load).
2. The average ultimate vertical load for a single hanger from all tests divided by six (where six or more tests are conducted).
3. The average from all tests of the vertical loads that produce a vertical movement of the joist with respect to the header is 0.125 inch (3.2 mm).
4. The sum of the allowable design loads for nails or other fasteners utilized to secure the joist hanger to the wood members and allowable bearing loads that contribute to the capacity of the hanger.
5. The allowable design load for the wood members forming the connection.

1711.1.3 Torsional moment capacity for joist hangers. The torsional moment capacity for the joist hanger shall be determined by testing at least three joist hanger assemblies as specified in ASTM D 1761. The allowable torsional moment of the joist hanger shall be the average torsional moment at which the lateral movement of the top or bottom of the joist with respect to the original position of the joist is 0.125 inch (3.2 mm).

1711.1.4 Design value modifications for joist hangers. Allowable design values for joist hangers that are determined by Item 4 or 5 in Section 1711.1.2 shall be permitted to be modified by the appropriate duration of loading factors as specified in AF&PA NDS but shall not exceed the direct loads as determined by Item 1, 2 or 3 in Section 1711.1.2. Allowable design values determined by Item 1, 2 or 3 in Section 1711.1.2 shall not be modified by duration of loading factors.

1711.2 Concrete and clay roof tiles. Concrete and clay roof tiles shall be tested in accordance with Sections 1711.2.1 and 1711.2.2.

1711.2.1 Overturning resistance. Concrete and clay roof tiles shall be tested to determine their resistance to overturning due to wind in accordance with ICC SBCCI SSTD 11 and Chapter 15.

1711.2.2 Wind tunnel testing. When roof tiles do not satisfy the limitations in Chapter 16 for rigid tile, a wind tunnel test shall be used to determine the wind characteristics of the concrete or clay tile roof covering in accordance with ICC SBCCI SSTD 11 and Chapter 15.